



# APC Training for PCLDPE

## Production Team Training Module

17<sup>th</sup> & 21<sup>st</sup> June 2021

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# APC Overview

Insert subtitles here

# APC Overview

## What is APC?

Advanced Process Control (APC) is an application with some features

- Multivariable – controls many variables in a single MPC controller
- Predictive - uses historical data to predict current and future state
- Control – calculates moves to be made to bring the process to desired condition and sends signals to regulatory controllers.
- Model-based – models describe interactions of variables in the APC envelope.
- Has optimization capability in which economic objectives can be achieved by MPC controller by maximizing or minimizing desired variables or keep them at preferred values.

# APC Overview

## Basic Terminologies:

- ❑ Independent Variable – Manipulated Variable (MV) – Disturbance Variable/Feedforward Variable (DV/FF)
- ❑ Dependent Variable – Controlled Variable (CV)
- ❑ There are 2 types of APC:
  1. Linear APC to handle linear processes

Most of APCs used in PETRONAS are linear. (SMOC, RMPCT, DMC+)

Concepts of :

- o Preservation of scale
- o Superposition

2. Non-linear APC • Used in PETLIN and MTBE

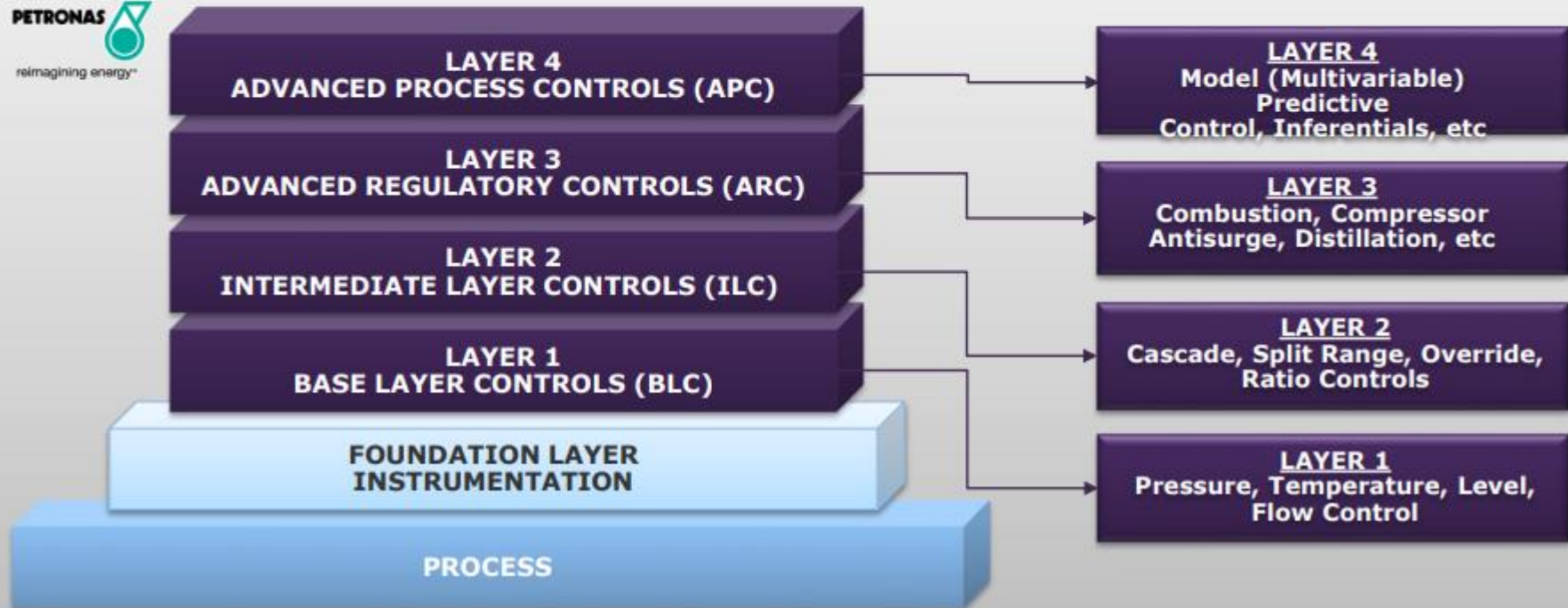
# APC Overview

## What is APC ?

According to SKG16.4 Introduction to MPC:

*" The field of (Advanced) Process Control refers to the application of appropriate control technology to achieve the followings:*

- *achieve greater process stability*
- *improved product specification*
- *increased yield and throughput*
- *increase efficiency"*



# Without APC



Automatic (SP)

Focus on Individual loops, flow, level, pressure, temperature

**Slower, Inconsistence Response time !**

Manual (CV)



DCS

SP

SP

SP

SP

SP

FC

TC

PC

SC

TC

LC

Operators need to memorize constraints and set Sp / Cv continuously

Manual optimization to meet product spec

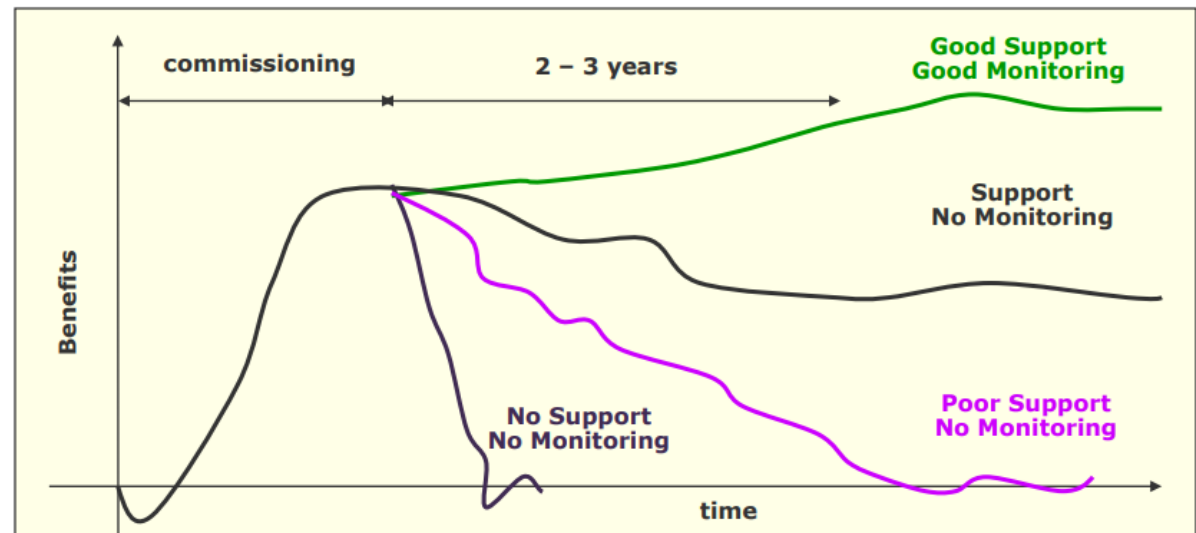
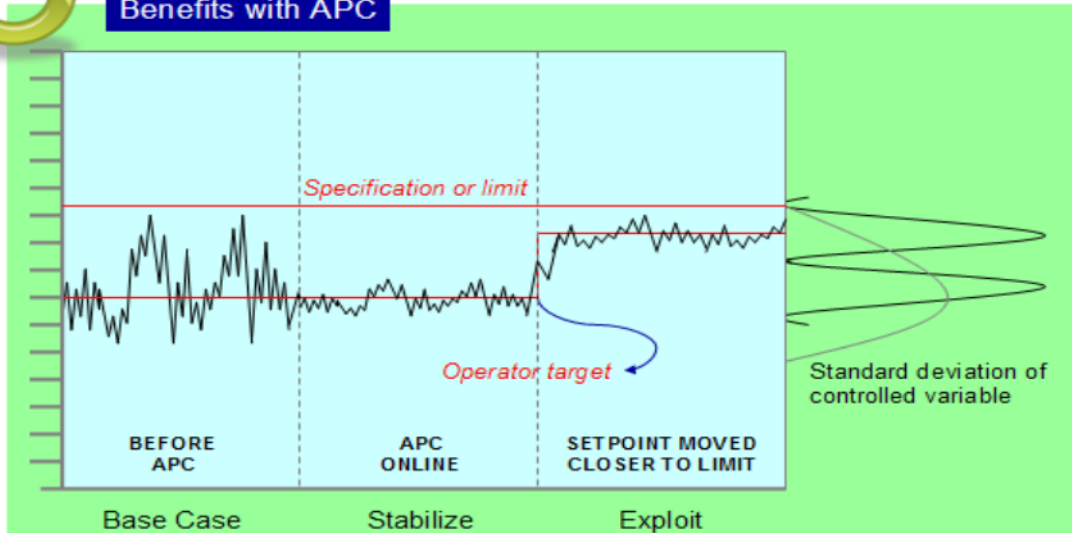
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# Benefits with APC



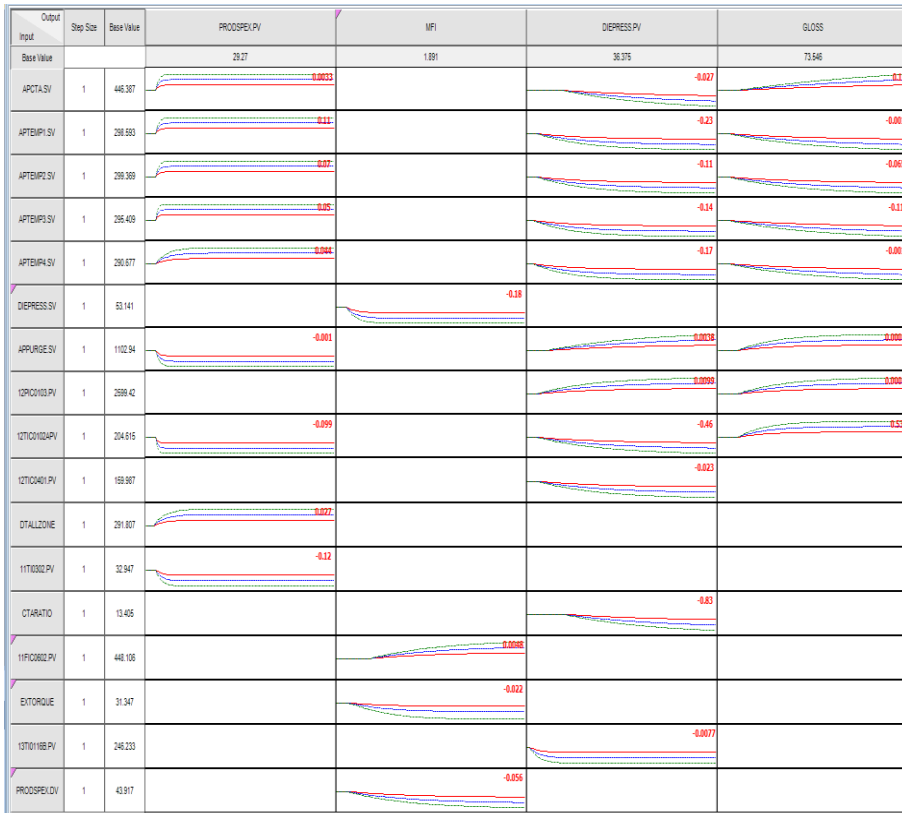
## Benefits with APC



# PCLDPE APC Overview

# PCLDPE APC Overview

How APC know , what MVs to be moved and how much to be moved?



| CV | Variables     | PRODSPEX.PV | MFI | DIEPRESS.PV | GLOSS |
|----|---------------|-------------|-----|-------------|-------|
| MV | APCTA.SV      |             |     | -           |       |
|    | APTEMP1.SV    | +           |     |             | -     |
|    | APTEMP2.SV    | +           |     |             | -     |
|    | APTEMP3.SV    | +           |     |             | -     |
|    | APTEMP4.SV    | +           |     |             | -     |
|    | DIEPRESS.SV   |             | -   |             |       |
|    | APPURGE.SV    | -           |     |             | +     |
| DV | 11FIC0401.SV  |             |     | +           |       |
|    | 12PIC0103.PV  |             |     | +           | +     |
|    | 12TIC0102A.PV | -           |     | -           | +     |
|    | 12TIC0401.PV  |             |     | -           |       |
|    | APTEMP1.SV    |             |     | -           |       |
|    | APTEMP2.SV    |             |     | -           |       |
|    | APTEMP3.SV    |             |     | -           |       |
|    | APTEMP4.SV    |             |     | -           |       |
|    | DTALLZONE     | +           |     |             |       |
|    | 11TIO302.PV   | -           |     |             |       |
|    | CTARATIO      |             |     | -           |       |
|    | 11FIC0602.PV  | +           | +   |             | +     |
|    | 13JIO101.PV   |             | -   |             |       |
|    | 13TIO116B.PV  |             |     | -           |       |
|    | PRODSPEX.PV   |             | -   |             |       |

There is relationship between all MVs with the CVs ,and it is called APC model.

# PCLDPE APC Overview

Advanced Process Control application was implemented in PCLDPE in order to achieve several objectives as below:

- ❑ Minimize variance in MFI and Gloss during steady state operation
- ❑ Minimize transition times by automating transitions
- ❑ Maximizing production rate without compromising quality
- ❑ Minimizing purge gas without compromising quality subject to constraints

These objectives can be achieved by having few APC components which are:

Polymer Inferential Properties (Aspen IQ)

Non-Linear Advanced Process Control

Process Recipe and Transition Manager (Grade Transition Manager)

# PCLDPE APC Overview

## Manipulated Variables

| No | Model Tag  | DCS Point   | Description      |
|----|------------|-------------|------------------|
| 1  | APCTA.SV   | APCTA.RSV   | Propylene Flow   |
| 2  | APTEMP1.SV | APTEMP1.RSV | Peak Temp Zone 1 |
| 3  | APTEMP2.SV | APTEMP2.RSV | Peak Temp Zone 2 |
| 4  | APTEMP3.SV | APTEMP3.RSV | Peak Temp Zone 3 |
| 5  | APTEMP4.SV | APTEMP4.RSV | Peak Temp Zone 4 |
| 6  | APPURGE.SV | APPURGE.RSV | Purge Gas Flow   |

## Controlled Variables

| No | Model Tag   | IP21 Point         | DCS Point         | Description        |
|----|-------------|--------------------|-------------------|--------------------|
| 1  | MFI         | IQ_MFI_BA          |                   | Melt Flow Index    |
| 2  | DIEPRESS.PV | IQ_DIEPRESS_BA .PV |                   | Die Plate Pressure |
| 3  | GLOSS       | IQ_GLOSS_BA        |                   | Gloss Lab          |
| 4  | PRODSPEX    |                    | PRODSPEX_PVFLT.PV | Production Rate    |

# PCLDPE APC Overview

Disturbance  
Variables

| No | Model Tag    | DCS Point           | Description                |
|----|--------------|---------------------|----------------------------|
| 1  | 12PIC0103SV  | 12PIC0103_PVFLT.PV  | Reactor Pressure           |
| 2  | 12TIC0401PV  | 12TIC0401_PVFLT.PV  | RHW Temp PV                |
| 3  | RXSTARTTEMP  | 12TIC0102A_PVFLT.PV | Rxn Start Temp (Preheater) |
| 4  | APTEMP1PV    | APTEMP1.RSV         | Peak Temp Zone 1           |
| 5  | APTEMP2PV    | APTEMP2.RSV         | Peak Temp Zone 2           |
| 6  | APTEMP3PV    | APTEMP3.RSV         | Peak Temp Zone 3           |
| 7  | APTEMP4PV    | APTEMP4.RSV         | Peak Temp Zone 4           |
| 8  | 11FIC0401.PV | APPURGE.RSV         | Purge Gas Flow             |
| 9  | CTARATIO     | CTARATIO_PVFLT.PV   | CTAFLOW/Ethylene Flow      |
| 10 | 13TI0116B.PV | 13TI0116B_PVFLT.PV  | Extruder Temperature       |
| 11 | 13JI0101.PV  | 13JI0101_PVFLT.PV   | Extruder Torque            |
| 12 | 11TI0302.PV  | 11TI0302_PVFLT.PV   | Suction temperature        |
| 13 | DTALLZONE    | DTALLZONE_PVFLT.PV  | Summation of all DIFFTEMP  |
| 14 | 11FIC0602.SV | 11FIC0602_PVFLT.PV  | CTA flow                   |
| 15 | PRODSPEX     | PRODSPEX_PVFLT.PV   | Production Rate            |

# Web Viewer Overview

| Application                                                      |                             | Last Run                                                                                                                            |                 |
|------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <div> <div></div> <div>PETLIN_ASV2020</div> <div>On</div> </div> |                             | <div> <div>11/29/2021 6:08:00 PM</div> <div>MISO Move Plan Calculation</div> <div>(1/29/2021 1:47:00 PM Controller On)</div> </div> |                 |
| Independents                                                     |                             | Filter                                                                                                                              |                 |
| Name                                                             | Description                 | Critical                                                                                                                            | Combined Status |
| AP...                                                            | FLOW                        | No                                                                                                                                  | Lo Limit        |
| AP...                                                            | 1                           | No                                                                                                                                  | Hi Limit        |
| AP...                                                            | 2                           | No                                                                                                                                  | Hi Limit        |
| APTEMP3.SV                                                       | REACTOR PEAK TEMP 3         | No                                                                                                                                  | Hi Limit        |
| APTEMP4.SV                                                       | REACTOR PEAK TEMP 4         | No                                                                                                                                  | Hi Limit        |
| DIEPRESS.SV*                                                     | 13A249 DIEPLATE PRESSURE    | No                                                                                                                                  | Hi Limit        |
| APPURGE.SV                                                       | S118 PUR...                 | No                                                                                                                                  | Hi Limit        |
| 12PIC0103.PV                                                     | REACTOR...                  | No                                                                                                                                  | Normal          |
| 12TIC0102APV                                                     | PREHEAT...                  | No                                                                                                                                  | Normal          |
| 12TIC0401.PV                                                     | R201 RHW SUPPLY TEMP        | No                                                                                                                                  | Normal          |
| DTALLZONE                                                        | PEAK VALLEY TOTAL TEMP DIFF | No                                                                                                                                  | Normal          |
| 11TI0302.PV                                                      | K103 SUCT TEMPERATURE       | No                                                                                                                                  | Normal          |
| CTARATIO                                                         | CTA/ETHYLENE                | No                                                                                                                                  | Normal          |
| 11FIC0602.PV*                                                    | PROPYLENE TOK102 FLOW       | No                                                                                                                                  | Normal          |
| EXTORQUE*                                                        | MAIN EXTRU TORQUE           | No                                                                                                                                  | Normal          |
| 13TI0116B.PV                                                     | 13A249 DIEPLATE TEMP        | No                                                                                                                                  | Normal          |
| PRODSPEX.DV*                                                     | PROD SPEED AFTER EXTRUDR    | No                                                                                                                                  | Normal          |
| Dependents                                                       |                             | Filter                                                                                                                              |                 |
| Name                                                             | Description                 | Critical                                                                                                                            | Combined Status |
| PRODSPEX.PV                                                      | PROD SPEED AFTER EXTRUDR    | No                                                                                                                                  | < Lo Limit      |
| MFI*                                                             | Melt Flow rate              | No                                                                                                                                  | Normal          |
| DIEPRESS.PV                                                      | MAIN EXT DIEPLATE PRESSURE  | No                                                                                                                                  | Hi Limit        |
| GLOSS                                                            | Gloss Line 1 - RUN          | No                                                                                                                                  | Normal          |

Switch ON button for PETLIN\_ASV2020

Switch ON button for MV

Desired CV limit operating range

Switch ON button for CV

Desired MV limit operating range

Calculated SS value by APC

External target. The desired target by operator/GTM

# Main Controlled Variables

## a) Melt Flow Index and Diepress

One of the most important quality to be controlled is MFI. Significant difference for this APC revamp is, MFI will be controlled by Diepress.SV.

Diepress.SV is a dummy setpoint which will provide an external target to another CV, Diepress.PV. Diepress.PV will manipulate CTA flow.

| Manipulated Variables (MV)                | Controlled Variables (CV)         | APC Controller Objectives                                                                                                                                                                                                                                       |
|-------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DIEPRESS.SV</b><br>(Dieplate Pressure) | MFI<br>(Melt Flow Index Analyser) | To control MFI at target (i.e. 5) as set by the operator or APR/TM. Main handle is die plate pressure. However, this is not a real controller. DIEPRESS.SV will be used as external target for DIEPRESS.PV.                                                     |
|                                           |                                   | MFI measurement is a predicted value based on unbiased prediction from an inferential model and bias update value. Mainly used all the time. MFI measurement is a predicted value based on unbiased prediction from an inferential model and bias update value. |
|                                           |                                   | Bias update is configured to receive input from 2 sources which are analyser (13AI0101) and lab measurement.                                                                                                                                                    |

| Manipulated Variables (MV)       | Controlled Variables (CV)           | APC Controller Objectives                                                                                               |
|----------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>APCTA</b><br>(Propylene Flow) | DIEPRESS.PV<br>(Die plate Pressure) | To control die plate pressure at its target and its target coming from DIEPRESS.SV (sent by APC)                        |
|                                  |                                     | Diepress measurement is a predicted value based on unbiased prediction from an inferential model and bias update value. |
|                                  |                                     | Bias update is configured to receive input only from the filtered online dieplate measurement 13PI0107B PVFLT.PV        |

# a) Melt Flow Index and Diepress

| Application      |                             | Last Run |                       |                                                                 |             |                    |             |                     |                    |              |
|------------------|-----------------------------|----------|-----------------------|-----------------------------------------------------------------|-------------|--------------------|-------------|---------------------|--------------------|--------------|
| ● PETLIN_ASV2020 |                             | On       | 11/29/2021 6:08:00 PM | MISO Move Plan Calculation (1/29/2021 1:47:00 PM Controller On) |             |                    |             |                     |                    |              |
| Independents     |                             | Filter   | None                  |                                                                 |             |                    |             |                     |                    | Copy         |
| Name             | Description                 | Critical | Combined Status       | Service Request                                                 | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Current Move |
| AP...            | ...                         | No       | Lo Limit              | On                                                              | On          | 440.000            | 494.817     | 550.000             | 440.000            | 1.670        |
| AP...            | ...                         | No       | Hi Limit              | On                                                              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| AP...            | ...                         | No       | Hi Limit              | On                                                              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP3.SV       | REACTOR PEAK TEMP 3         | No       | Hi Limit              | On                                                              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP4.SV       | REACTOR PEAK TEMP 4         | No       | Hi Limit              | On                                                              | On          | 294.000            | 295.000     | 295.000             | 295.000            | 0.000        |
| DIEPRESS.SV*     | 13A249 DIEPLATE PRESSURE    | No       | Lo Limit              | On                                                              | On          | 47.000             | 47.011      | 49.000              | 47.000             | 0.000        |
| APPURGE.SV       | S118 PURGE GAS FLOW         | No       | Lo Limit              | On                                                              | On          | 850.000            | 850.000     | 1200.000            | 850.000            | 0.000        |
| 12PIC0103.PV     | REACTOR PRESSURE            | No       | Normal                |                                                                 |             |                    | 2554.594    |                     | 2554.594           |              |
| 12TIC0102APV     | PREHEATER 2 TEMP            | No       | Normal                |                                                                 |             |                    | 167.952     |                     | 167.952            |              |
| 12TIC0401.PV     | R201 RHW SUPPLY TEMP        | No       | Normal                |                                                                 |             |                    | 169.944     |                     | 169.944            |              |
| DTALLZONE        | PEAK VALLEY TOTAL TEMP DIFF | No       | Normal                |                                                                 |             |                    | 270.478     |                     | 270.478            |              |
| 11TI0302.PV      | K103 SUCT TEMPERATURE       | No       | Normal                |                                                                 |             |                    | 33.935      |                     | 33.935             |              |
| CTARATIO         | CTA/ETHYLENE                | No       | Normal                |                                                                 |             |                    | 15.965      |                     | 15.965             |              |
| 11FIC0602.PV*    | PROPYLENE TOK102 FLOW       | No       | Normal                | On                                                              |             |                    | 491.780     |                     | 491.780            |              |
| EXTORQUE*        | MAIN EXTRU TORQUE           | No       | Normal                |                                                                 |             |                    | 27.833      |                     | 27.833             |              |
| 13TI0116B.PV     | 13A249 DIEPLATE TEMP        | No       | Normal                |                                                                 |             |                    | 243.282     |                     | 243.282            |              |
| PRODSPEX.DV*     | PROD SPEED AFTER EXTRUDR    | No       | Normal                |                                                                 |             |                    | 30.707      |                     | 30.707             |              |
| Dependents       |                             | Filter   | None                  |                                                                 |             |                    |             |                     |                    | Copy         |
| Name             | Description                 | Critical | Combined Status       | Service Request                                                 | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Target       |
| PRODSPEX.PV      | PROD SPEED AFTER EXTRUDR    | No       | < Lo Limit            | On                                                              |             | 32.000             | 30.707      | 36.000              | 30.836             |              |
| MFI*             | Melt Flow rate              | No       | Normal                | On                                                              |             | 4.600              | 4.689       | 5.700               | 5.653              | 5.100        |
| DIEPRESS.PV      | MAIN EXT DIEPLATE PRESSURE  | No       | Normal                | On                                                              |             | 47.000             | 47.952      |                     | 45.755             | 47.011       |
| GLOSS            | Gloss Line 1 - RUN          | No       | Normal                | On                                                              |             | 62.000             | 70.54       |                     | 77.79              |              |

Switch ON button for PETLIN\_ASV2020

High and Low limit for its MV. It will be changed based on product grade either by operator or GTM

High and Low limit for MV will be similar with high and low limit for CV

Switch ON button for MFI CV

External target for MFI. Operator will key in the desired MFI target here. Source of inputs, either operator or GTM.

# a) Melt Flow Index and Diepress

Application

Last Run

PETLIN\_ASV2020

On

On

1/29/2021 6:08:00 PM

MISO Move Plan Calculation

(1/29/2021 1:47:00 PM Controller On)

Independents

Filter

None

Copy

| Name          | Description                 | Critical | Combined Status | Service Request | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Current Move |
|---------------|-----------------------------|----------|-----------------|-----------------|-------------|--------------------|-------------|---------------------|--------------------|--------------|
| APCTA.SV      | PROPYLENE TOK102 FLOW       | No       | Lo Limit        | On              | On          | 440.000            | 494.817     | 550.000             | 440.000            | 1.670        |
| APTEMP1.SV    | REACTOR PEAK TEMP 1         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP2.SV    | REACTOR PEAK TEMP 2         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP3.SV    | REACTOR PEAK TEMP 3         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP4.SV    | REACTOR PEAK TEMP 4         | No       | Hi Limit        | On              | On          | 294.000            | 295.000     | 295.000             | 295.000            | 0.000        |
| DIEPRESS.SV*  | 13A249 DIEPLATE PRESSURE    | No       | Lo Limit        | On              | On          | 47.000             | 47.011      | 49.000              | 47.000             | 0.000        |
| APPURGE.SV    | S118 PURGE GAS FLOW         | No       | Lo Limit        | On              |             | 850.000            | 850.000     | 1200.000            | 850.000            | 0.000        |
| 12PIC0103.PV  | REACTOR PRESSURE            | No       | Normal          | On              |             |                    | 2554.594    |                     | 2554.594           |              |
| 12TIC0102APV  | PREHEATER 2 TEMP            | No       | Normal          | On              |             |                    |             |                     | 167.952            |              |
| 12TIC0401.PV  | R201 RHW SUPPLY TEMP        | No       | Normal          | On              |             |                    |             |                     | 169.944            |              |
| DTALLZONE     | PEAK VALLEY TOTAL TEMP DIFF | No       | Normal          | On              |             |                    |             |                     | 270.478            |              |
| 11TI0302.PV   | K103 SUCT TEMPERATURE       | No       | Normal          | On              |             |                    |             |                     | 33.935             |              |
| CTARATIO      | CTA/ETHYLENE                | No       | Normal          | On              |             |                    |             |                     | 15.965             |              |
| 11FIC0602.PV* | PROPYLENE TOK102 FLOW       | No       | Normal          | On              |             |                    | 491.780     |                     | 491.780            |              |
| EXTORQUE*     | MAIN EXTRU TORQUE           | No       | Normal          |                 |             |                    | 27.833      |                     | 27.833             |              |
| 13TI0116B.PV  | 13A249 DIEPLATE TEMP        | No       | Normal          |                 |             |                    | 243.282     |                     | 243.282            |              |
| PRODSPEX.DV*  | PROD SPEED AFTER EXTRUDR    | No       | Normal          |                 |             |                    | 30.707      |                     | 30.707             |              |

Dependents

Filter

None

Copy

| Name        | Description                | Critical | Combined Status | Service Request | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Target |
|-------------|----------------------------|----------|-----------------|-----------------|-------------|--------------------|-------------|---------------------|--------------------|--------|
| PRODSPEX.PV | PROD SPEED AFTER EXTRUDR   | No       | < Lo Limit      | On              |             | 32.000             | 30.707      | 36.000              | 30.836             |        |
| MFI*        | Melt Flow rate             | No       | Normal          | On              |             | 4.600              | 4.689       | 5.700               | 5.653              | 5.100  |
| DIEPRESS.PV | MAIN EXT DIEPLATE PRESSURE | No       | < Lo Limit      | On              |             | 47.000             | 47.952      | 49.000              | 45.765             | 47.011 |
| GLOSS       | Gloss Line 1 - RUN         | No       | Normal          | On              |             | 62.000             | 70.548      | 70.000              | 75.779             |        |

Ensure Diepress.SV service request is on every time APC is controlling MFI

High and Low limit for MV will be similar with high and low limit for CV

Ensure Diepress.SV service request is on every time APC is controlling MFI

High and Low limit for MV will be similar with high and low limit for CV

External target for Diepress.PV is coming from Diepress.SV

## b) Gloss

Another quality to be controlled is Gloss. The quality will be controlled by Peak Temperature 1 / 2 / 3 / 4 and purge gas flow. Purge gas flow is configured to be minimized all the time.

| Manipulated Variables (MV)           | Controlled Variables (CV) | APC Controller Objectives                                                                                   |
|--------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>APTEMP1</b><br>(Peak Temp Zone 1) | GLOSS<br>(Gloss)          | To control Gloss within limit as set by the operator or APR/TM. Main handle is Peak temperature.            |
| <b>APTEMP2</b><br>(Peak Temp Zone 2) |                           | GLOSS is predicted value from inferential model. It is not used during high MFI (M022X & M050X) production. |
| <b>APTEMP3</b><br>(Peak Temp Zone 3) |                           |                                                                                                             |
| <b>APTEMP4</b><br>(Peak Temp Zone 4) |                           |                                                                                                             |
| <b>11FIC0401</b><br>(Purge gas flow) |                           | During gloss low limit violation, purge gas will be maximized and peak temperatures will be reduced         |

## b) Gloss

Application

Last Run

PETLIN\_ASV2020

On

On

1/29/2021 6:08:00 PM

MISO Move Plan Calculation

(1/29/2021 1:47:00 PM Controller On)

Independents

Filter

None

Copy

| Name          | Description                 | Critical | Combined Status | Service Request | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Current Move |
|---------------|-----------------------------|----------|-----------------|-----------------|-------------|--------------------|-------------|---------------------|--------------------|--------------|
| APCTA.SV      | PROPYLENE TOK102 FLOW       | No       | Lo Limit        | On              | On          | 440.000            | 494.817     | 550.000             | 440.000            | 1.670        |
| APTEMP1.SV    | REACTOR PEAK TEMP 1         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP2.SV    | REACTOR PEAK TEMP 2         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP3.SV    | REACTOR PEAK TEMP 3         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP4.SV    | REACTOR PEAK TEMP 4         | No       | Hi Limit        | On              | On          | 294.000            | 295.000     | 295.000             | 295.000            | 0.000        |
| DIEPRESS.SV*  | 13A249 DIEPLATE PRESSURE    | No       | Lo Limit        | On              | On          | 47.000             | 47.011      | 49.000              | 47.000             | 0.000        |
| APPURGE.SV    | S118 PURGE GAS FLOW         | No       | Lo Limit        | On              | On          | 850.000            | 850.000     | 1200.000            | 850.000            | 0.000        |
| 12PIC0103.PV  | REACTOR PRESSURE            | No       | Normal          | On              |             |                    | 2554.594    |                     | 2554.594           |              |
| 12TIC0102APV  | PREHEATER 2 TEMP            | No       | Normal          | On              |             |                    | 167.952     |                     | 167.952            |              |
| 12TIC0401.PV  | R201 RHW SUPPLY TEMP        | No       | Normal          | On              |             |                    | 169.944     |                     | 169.944            |              |
| DTALLZONE     | PEAK VALLEY TOTAL TEMP DIFF | No       | Normal          | On              |             |                    | 270.478     |                     | 270.478            |              |
| 11TI0302.PV   | K103 SUCT TEMPERATURE       | No       | Normal          | On              |             |                    | 33.935      |                     | 33.935             |              |
| CTARATIO      | CTA/ETHYLENE                | No       | Normal          | On              |             |                    | 15.965      |                     | 15.965             |              |
| 11FIC0602.PV* | PROPYLENE TOK102 FLOW       | No       | Normal          | On              |             |                    | 491.780     |                     | 491.780            |              |
| EXTORQUE*     | MAIN EXTRU TORQUE           | No       | Normal          | On              |             |                    | 27.833      |                     | 27.833             |              |
| 13TI0116B.PV  | 13A249 DIEPLATE TEMP        | No       | Normal          | On              |             |                    | 243.282     |                     | 243.282            |              |
| PRODSPEX.DV*  | PROD SPEED AFTER EXTRUDR    | No       | Normal          | On              |             |                    | 30.707      |                     | 30.707             |              |

Dependents

Filter

None

Copy

| Name        | Description                | Critical | Combined Status | Service Request | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Target |
|-------------|----------------------------|----------|-----------------|-----------------|--------------------|-------------|---------------------|--------------------|--------|
| PRODSPEX.PV | PROD SPEED AFTER EXTRUDR   | No       | < Lo Limit      | On              | 32.000             | 30.707      | 36.000              | 30.836             |        |
| MFI*        | Melt Flow rate             | No       | Normal          | On              | 4.600              | 4.689       | 5.700               | 5.653              | 5.100  |
| DIEPRESS.PV | MAIN EXT DIEPLATE PRESSURE | No       | < Lo Limit      | On              | 47.000             | 47.952      | 49.000              | 45.765             | 47.011 |
| GLOSS       | Gloss Line 1 - RUN         | No       | Normal          | On              | 62.000             | 70.548      | 78.000              | 75.779             |        |

For product grade without gloss quality, this CV will be turned off automatically by GTM

Gloss will be controlled between low and high limit configured by panel operator or GTM

## c) Prodspex

Prodspex is controlled by Peak Temperature 1 / 2 / 3 / 4 and purge gas flow where all the peak temperature are configured to be maximized and purge gas flow is configured to be minimized to maximize production

| Manipulated Variables (MV)              | Controlled Variables (CV)     | APC Controller Objectives                                                                                                                                                                                                                                        |
|-----------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>APTEMP1</b><br>(Peak Temp Zone 1)    | PRODSPEX<br>(Production Rate) | To maximize production rate to its high limit as set by the operator or APR/TM. Main handle is Peak temperature.                                                                                                                                                 |
| <b>APTEMP2</b><br>(Peak Temp Zone 2)    |                               | PRODSPEX is used all the time to optimize the production rate.                                                                                                                                                                                                   |
| <b>APTEMP3</b><br>(Peak Temp Zone 3)    |                               |                                                                                                                                                                                                                                                                  |
| <b>APTEMP4</b><br>(Peak Temp Zone 4)    |                               | Purge gas will be set to be minimized to maximize Prodspex. During gloss low limit violation, purge gas and peak will be directly responding to gloss measurement due to lower SS Low rank (means higher priority) parameter configured for gloss than prodspex. |
| <b>11FIC0401.PV</b><br>(Purge gas flow) |                               | This is configured due to both CV (gloss & prodspex) are utilizing the same handles                                                                                                                                                                              |

## c) Prodsplex

Application

Last Run

PETLIN\_ASV2020

On

On

1/29/2021 6:08:00 PM

MISO Move Plan Calculation

(1/29/2021 1:47:00 PM Controller On)

Independents

Filter

None

Copy

| Name          | Description                 | Critical | Combined Status | Service Request | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Current Move |
|---------------|-----------------------------|----------|-----------------|-----------------|-------------|--------------------|-------------|---------------------|--------------------|--------------|
| APCTA.SV      | PROPYLENE TOK102 FLOW       | No       | Lo Limit        | On              | On          | 440.000            | 494.817     | 550.000             | 440.000            | 1.670        |
| APTEMP1.SV    | REACTOR PEAK TEMP 1         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP2.SV    | REACTOR PEAK TEMP 2         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP3.SV    | REACTOR PEAK TEMP 3         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP4.SV    | REACTOR PEAK TEMP 4         | No       | Hi Limit        | On              | On          | 294.000            | 295.000     | 295.000             | 295.000            | 0.000        |
| DIEPRESS.SV*  | 13A249 DIEPLATE PRESSURE    | No       | Lo Limit        | On              | On          | 47.000             | 47.011      | 49.000              | 47.000             | 0.000        |
| APPURGE.SV    | S118 PURGE GAS FLOW         | No       | Lo Limit        | On              | On          | 850.000            | 850.000     | 1200.000            | 850.000            | 0.000        |
| 12PIC0103.PV  | REACTOR PRESSURE            | No       | Normal          | On              |             |                    | 2554.594    |                     | 2554.594           |              |
| 12TIC0102APV  | PREHEATER 2 TEMP            | No       | Normal          | On              |             |                    | 167.952     |                     | 167.952            |              |
| 12TIC0401.PV  | R201 RHW SUPPLY TEMP        | No       | Normal          | On              |             |                    | 169.944     |                     | 169.944            |              |
| DTALLZONE     | PEAK VALLEY TOTAL TEMP DIFF | No       | Normal          | On              |             |                    | 270.478     |                     | 270.478            |              |
| 11TI0302.PV   | K103 SUCT TEMPERATURE       | No       | Normal          | On              |             |                    | 33.935      |                     | 33.935             |              |
| CTARATIO      | CTA/ETHYLENE                | No       | Normal          | On              |             |                    | 15.965      |                     | 15.965             |              |
| 11FIC0602.PV* | PROPYLENE TOK102 FLOW       | No       | Normal          | On              |             |                    | 491.780     |                     | 491.780            |              |
| EXTORQUE*     | MAIN EXTRU TORQUE           | No       | Normal          | On              |             |                    | 27.833      |                     | 27.833             |              |
| 13TI0116B.PV  | 13A249 DIEPLATE TEMP        | No       | Normal          | On              |             |                    | 243.282     |                     | 243.282            |              |
| PRODSPLEX.DV* | PROD SPEED AFTER EXTRUDR    | No       | Normal          | On              |             |                    | 30.707      |                     | 30.707             |              |

Dependents

Filter

None

Copy

| Name         | Description                | Critical | Combined Status | Service Request | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Target |
|--------------|----------------------------|----------|-----------------|-----------------|--------------------|-------------|---------------------|--------------------|--------|
| PRODSPLEX.PV | PROD SPEED AFTER EXTRUDR   | No       | < Lo Limit      | On              | 32.000             | 30.707      | 36.000              | 30.836             |        |
| MFI*         | Melt Flow rate             | No       | Normal          | On              | 4.600              | 4.689       | 5.700               | 5.653              | 5.100  |
| DIEPRESS.PV  | MAIN EXT DIEPLATE PRESSURE | No       | < Lo Limit      | On              | 47.000             | 47.952      | 49.000              | 45.765             | 47.011 |
| GLOSS        | Gloss Line 1 - RUN         | No       | Normal          | On              | 62.000             | 70.541      | 78.000              | 75.779             |        |

Prodsplex will be turned ON all the time to ensure production maximization.

Prodsplex will be controlled between low and high limit . APC will try to bring to high limit.

# How to operate APC

## Checklist before turning ON APC.

- ☐ Plant is running at normal and stable condition.
- ☐ Check and verify lab readings for MFI & Gloss.
- ☐ Inferential is updated with latest APC QC Lab result for MFI & Gloss.
- ☐ Ensure that you know what quality you are targeting for. (i.e. MFI for specific grade)
- ☐ Check the mode of your MVs. Ensure that it is in Auto mode. (APTEMP1, APCTA, APPURGE etc)
- ☐ Check your MVs and CVs Operating Limits on PCWS. Make sure you know where are you operating.
- ☐ Make sure you know the operating limits (Product Recipe) and what do you want to target for before you turn on the controller
- ☐ Ensure Diepress.SV and Diepress.PV are turned ON to allow MFI is controlled by APC.
- ☐ Ensure the high and low limit of Diepress.PV is within the recommended limit. Please refer to table on [page 28](#).
- ☐ Ensure alarm annunciator for watchdog has been acknowledged (if there is) before turning ON APC.
- ☐ Ensure MFI and gloss measurement are within the APC limit. Please note that if the step is not done, it will drive the controller to wrong direction.

# How to operate APC

## How to turn ON APC

1. Check MFI/Gloss reading. Compare with reference reading (Lab/ Analyzer). Check process operating condition and operating limit for MVs and CVs.
2. Upload Lab result if required to update the inferential. It is advised to turn ON APC only after lab bias is updated
3. Turn ON PETLIN\_ASV2020. Watchdog will reset automatically.
4. Adjust MV lower and upper operating limit
5. Adjust CV lower and upper operating limit
6. Turn ON MV switch.
7. Turn ON CV switch.
8. Key in Target for MFI.

Application Last Run

● PETLIN\_ASV2020 On 1/29/2021 6:08:00 PM MISO Move Plan Calculation (1/29/2021 1:47:00 PM Controller On)

Independents

3

| Name          | Description                 | Critical | Combined Status | Service Request | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Current Move |
|---------------|-----------------------------|----------|-----------------|-----------------|-------------|--------------------|-------------|---------------------|--------------------|--------------|
| APCTA.SV      | PROPYLENE TOK102 FLOW       | No       | Lo Limit        | On              | On          | 440.000            | 494.817     | 550.000             | 440.000            | 1.670        |
| APTEMP1.SV    | REACTOR PEAK TEMP 1         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP2.SV    | REACTOR PEAK TEMP 2         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP3.SV    | REACTOR PEAK TEMP 3         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP4.SV    | REACTOR PEAK TEMP 4         | No       | Hi Limit        | On              | On          | 294.000            | 295.000     | 295.000             | 295.000            | 0.000        |
| DIEPRESS.SV*  | 13A249 DIEPLATE PRESSURE    | No       | Lo Limit        | On              | On          | 47.000             | 47.011      | 49.000              | 47.000             | 0.000        |
| APPURGE.SV    | S118 PURGE GAS FLOW         | No       | Lo Limit        | On              | On          | 850.000            | 850.000     | 1200.000            | 850.000            | 0.000        |
| 12PIC0103.PV  | REACTOR PRESSURE            | No       | Normal          | On              |             |                    | 2554.594    |                     | 2554.594           |              |
| 12TIC0102APV  | PREHEATER 2 TEMP            | No       | Normal          | On              |             |                    | 167.952     |                     | 167.952            |              |
| 12TIC0401.PV  | R201 RHW SUPPLY TEMP        | No       | Normal          | On              |             |                    | 169.944     |                     | 169.944            |              |
| DTALLZONE     | PEAK VALLEY TOTAL TEMP DIFF | No       | Normal          | On              |             |                    | 270.478     |                     | 270.478            |              |
| 11T0302.PV    | K103 SUCT TEMPERATURE       | No       | Normal          | On              |             |                    | 33.935      |                     | 33.935             |              |
| CTARATIO      | CTA/ETHYLENE                | No       | Normal          | On              |             |                    | 15.965      |                     | 15.965             |              |
| 11FIC0602.PV* | PROPYLENE TOK102 FLOW       | No       | Normal          | On              |             |                    | 491.780     |                     | 491.780            |              |
| EXTORQUE*     | MAIN EXTRU TORQUE           | No       | Normal          | On              |             |                    | 27.833      |                     | 27.833             |              |
| 13T0116B.PV   | 13A249 DIEPLATE TEMP        | No       | Normal          | On              |             |                    | 243.282     |                     | 243.282            |              |
| PRODSPEX.DV*  | PROD SPEED AFTER EXTRUDR    | No       | Normal          | On              |             |                    | 30.707      |                     | 30.707             |              |

Dependents

Filter

None

4

| Name        | Description                | Critical | Combined Status | Service Request | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Target |
|-------------|----------------------------|----------|-----------------|-----------------|-------------|--------------------|-------------|---------------------|--------------------|--------|
| PRODSPEX.PV | PROD SPEED AFTER EXTRUDR   | No       | < Lo Limit      | On              | On          | 32.000             | 30.707      | 36.000              | 30.836             |        |
| MFI*        | Melt Flow rate             | No       | Normal          | On              |             | 4.600              | 4.689       | 5.700               |                    | 5.100  |
| DIEPRESS.PV | MAIN EXT DIEPLATE PRESSURE | No       | < Lo Limit      | On              |             | 47.000             | 47.952      | 49.000              |                    | 47.011 |
| GLOSS       | Gloss Line 1 - RUN         | No       | Normal          | On              |             | 62.000             | 70.548      | 78.000              |                    |        |

5

7

8

# How to operate APC

## How to turn OFF APC

1. Turn OFF all MVs via service request
2. Turn OFF APC on PCWS

Application

PETLIN\_ASV2020

On

On

Last Run

1/29/2021 6:08:00 PM

MISO Move Plan Calculation

(1/29/2021 1:47:00 PM Controller On)

Independents

2

one

Copy

| Name          | Description                 | Critical | Combined Status | Service Request | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Current Move |
|---------------|-----------------------------|----------|-----------------|-----------------|-------------|--------------------|-------------|---------------------|--------------------|--------------|
| APCTA_SV      | PROPYLENE TOK102 FLOW       | No       | Lo Limit        | On              | On          | 440.000            | 494.817     | 550.000             | 440.000            | 1.670        |
| APTEMP1_SV    | REACTOR PEAK TEMP 1         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP2_SV    | REACTOR PEAK TEMP 2         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP3_SV    | REACTOR PEAK TEMP 3         | No       | Hi Limit        | On              | On          | 297.000            | 298.000     | 298.000             | 298.000            | 0.000        |
| APTEMP4_SV    | REACTOR PEAK TEMP 4         | No       | Hi Limit        | On              | On          | 294.000            | 295.000     | 295.000             | 295.000            | 0.000        |
| DIEPRESS_SV*  | 13A249 DIEPLATE PRESSURE    | No       | Lo Limit        | On              | On          | 47.000             | 47.011      | 49.000              | 47.000             | 0.000        |
| APPURGE_SV    | S118 PURGE GAS FLOW         | No       | Lo Limit        | On              | On          | 850.000            | 850.000     | 1200.000            | 850.000            | 0.000        |
| 12PIC0103_PV  | REACTOR PRESSURE            | No       | Normal          | On              |             |                    | 2554.594    |                     | 2554.594           |              |
| 12TIC0102APV  | PREHEATER 2 TEMP            | No       | Normal          | On              |             |                    | 167.952     |                     | 167.952            |              |
| 12TIC0401_PV  | R201 RHW SUPPLY TEMP        | No       | Normal          | On              |             |                    | 169.944     |                     | 169.944            |              |
| DTALLZONE     | PEAK VALLEY TOTAL TEMP DIFF | No       | Normal          | On              |             |                    | 270.478     |                     | 270.478            |              |
| 11TIO302_PV   | K103 SUCT TEMPERATURE       | No       | Normal          | On              |             |                    | 33.935      |                     | 33.935             |              |
| CTARATIO      | CTA/ETHYLENE                | No       | Normal          | On              |             |                    | 15.965      |                     | 15.965             |              |
| 11FIC0602_PV* | PROPYLENE TOK102 FLOW       | No       | Normal          | On              |             |                    | 491.780     |                     | 491.780            |              |
| EXTORQUE*     | MAIN EXTRU TORQUE           | No       | Normal          | On              |             |                    | 27.833      |                     | 27.833             |              |
| 13TIO116B_PV  | 13A249 DIEPLATE TEMP        | No       | Normal          | On              |             |                    | 243.282     |                     | 243.282            |              |
| PRODSPEXDV*   | PROD SPEED AFTER EXTRUDR    | No       | Normal          | On              |             |                    | 30.707      |                     | 30.707             |              |

Dependents

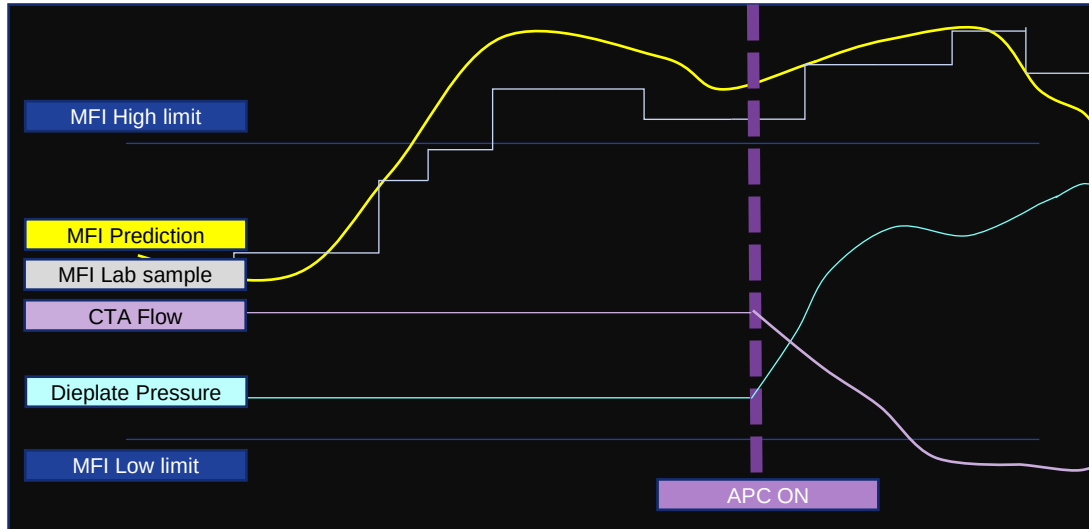
Filter

None

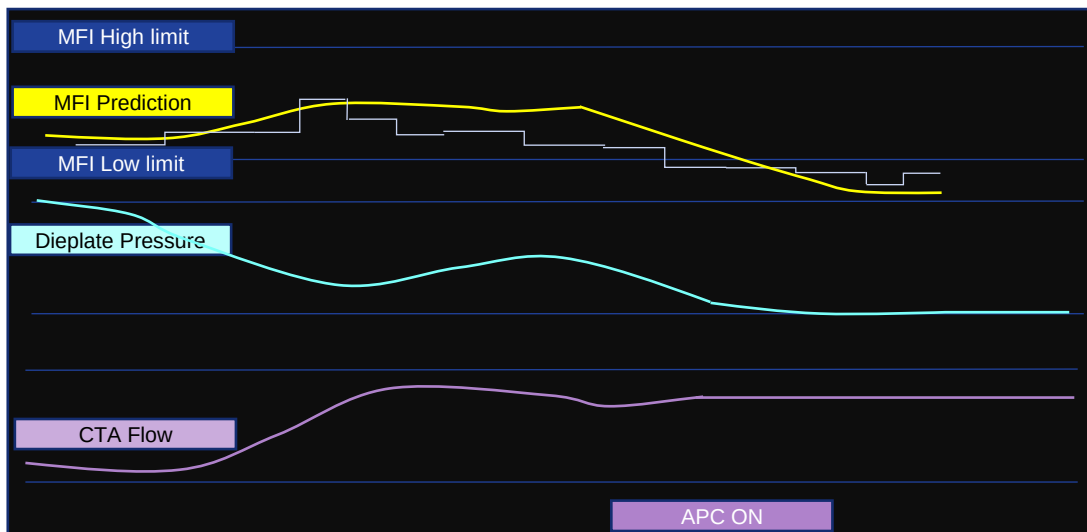
Copy

| Name        | Description                | Critical | Combined Status | Service Request | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Target |
|-------------|----------------------------|----------|-----------------|-----------------|--------------------|-------------|---------------------|--------------------|--------|
| PRODSPEX_PV | PROD SPEED AFTER EXTRUDR   | No       | < Lo Limit      | On              | 32.000             | 30.707      | 36.000              | 30.836             |        |
| MFI*        | Melt Flow rate             | No       | Normal          | On              | 4.600              | 4.689       | 5.700               | 5.653              | 5.100  |
| DIEPRESS_PV | MAIN EXT DIEPLATE PRESSURE | No       | < Lo Limit      | On              | 47.000             | 47.952      | 49.000              | 45.765             | 47.011 |
| GLOSS       | Gloss Line 1 - RUN         | No       | Normal          | On              | 62.000             | 70.548      | 78.000              | 75.779             |        |

How turning ON APC when MFI not in APC limit could destabilize the process



How CTA movement is restricted due to constrained dieplate pressure



**TRIVIAL  
QUESTION:**

If this happens,  
what should be  
done? Why?

# Diepress Low and High Limit

| Grade | Lo Limit | High Limit | Remarks                                                     |
|-------|----------|------------|-------------------------------------------------------------|
| N103X | 69.3     | 70.5       | Best Dieplate range* to attain and average mfi range of 0.3 |
| N109X | 59.19    | 62.20      | Dieplate Range - Historical Data 2020                       |
| N125Y | 54.5     | 56.5       | Best Dieplate range* to attain and average mfi range of 2.5 |
| C170X | 49.29    | 53.35      | Dieplate Range - Historical Data 2020                       |
| C150Y | 46.5     | 48.00      | Best Dieplate range* to attain and average mfi range of 5.0 |
| M022X | 37.5     | 39.00      | Best Dieplate range* to attain and average mfi range of 22  |
| M050X | 26.81    | 29.64      | Best Dieplate range* to attain and average mfi range of 50  |

\* Dieplate range is observed to vary according to the production rate running even of the same grade. A higher production rate will observe a higher dieplate range even at the same MFI value.

# APC / DCS Internal Logic & Calculation

# APC Internal Logic Calculation

There are few logics has been formulated in the calculation section. Basically, these logics are made in order to ensure smooth APC operation once commissioned. Below are some examples of logics configured

1) APTEMP4 MODE: This logic is to switch off MV APTEMP4 whenever peak temperature 4 is taken over by the override controller TIC0333A

```
IF MASTER_ON_REQ = 1 AND MASTER_ON_STAT = 1 THEN  
  
  IF TYA0950BPV =1 OR TYA0930BPV =1 THEN  
    SERV_REQ = 0  
  
  END IF  
  
END IF
```

# APC Internal Logic Calculation

2) APCTA\_PVTRACK: This logic is to ensure, whenever APC is OFF, the measurement will follow the current value. Thus, there is no bump whenever APC is turned ON. The same logic is duplicated for other MVs (APTEMP1/2/3/4\_PVTRACK, DIEPRESS\_PVTRACK, APPURGE\_PVTRACK) and CV target (MFI\_TARGET\_TRACK, DIE\_TARGET\_TRACK)

```
IF MASTER_ON_REQ = 0 or MASTER_ON_STAT = 0 or SRVREQ=0 THEN  
MEAS=PVFLT  
ELSE  
MEAS=PV  
END IF
```

# APC Internal Logic Calculation

3) APTEMP1\_AWC: This logic is to prevent huge difference between current DCS value to the setpoint given by APC. When the difference greater than the factor configured, APC will restrict the MV setpoint from increase or decrease by changing the status at AWS. The same logic has been configured for other MVs. The factor can be changed by APC engineer through PCWS and it is in engineering unit.

The example below is for DIFFTEMP

```
IF ABS(RSV-PVFLT) > DIFFTEMP THEN  
AWC=3  
END IF
```

| Variables     | AWC factor in PCWS |
|---------------|--------------------|
| APCTA         | 20 kg/hr           |
| APTEMP1/2/3/4 | 3 degC             |
| APPURGE       | 25 kg/hour         |

# DCS Internal Logic Calculation

## 4) APC Wind Up:

To tell APC controllers the status of PID controllers when its output reaches 0% or 100%. Final element fully close or fully open. (i.e. 11FIC0602.MV, 12TIC0720.MV). In the logic as per snapshot below, the trigger limit is the high and low clamp (MHI,MLO)

The status of Anti Wind-up is displayed on DCS and Web Viewer

```
LDPE\FCS0101\DR0027\APPURGE_AWC.edf

General-Purpose Calculation Formula (APPURGE_AWC

Line Number  General-Purpose Calculation Formula
001          PROGRAM
002
003          IF ({11FIC0401.ALARM.MHI}) THEN
004              CPV=2
005          ELSE IF ({11FIC0401.ALARM.MLO}) THEN
006              CPV=1
007          ELSE IF ({11FIC0401.ALARM.NR}) THEN
008              CPV=0
009          END IF
010
011          END
```

# DCS Internal Logic Calculation

## 5) Shed Mode:

The shed mode logic is required to return all of the PID controllers to their normal modes (e.g., “AUTO” for CTA and Peak Temperature controls) when the APC shuts down or when an individual MV is taken out of control. If the APC is turned ON, the DCS logic must change the modes of the PID controllers whose MVs are required to be ON, to the required DCS mode (CAS).

| No. | Tag name>Data item | Data | RuleNo.<br>StepNo.<br>Reference Signal Comment | 1      | 2      | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | Comment                 |
|-----|--------------------|------|------------------------------------------------|--------|--------|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------------------------|
| C01 | 4SG07002.FV        | 1    |                                                | M      | M      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | ON REQUEST APC          |
| C02 | 4SG07002.FV        | 0    |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | APC OFF MODE            |
| C03 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C04 | APPURGE.MV         | 0    | APPURGE MANUAL MODE                            | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | SERVICE OFF REQ APC/DCS |
| C05 | APPURGE.MV         | 1    | APPURGE MANUAL MODE                            | M      | M      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | SERVICE ON REQ APC      |
| C06 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C07 | 11FIC0401.MODE     | MAN  | 11S PURGE GAS FLOW                             | NNNNNN | NNNNNN | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C08 | 4SG07002.FV        | ON   |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | APC/LO 180 SEC TIMER UP |
| C09 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C10 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C11 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C12 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C13 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C14 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C15 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C16 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C17 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C18 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C19 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C20 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C21 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C22 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C23 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C24 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C25 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C26 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C27 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C28 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C29 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C30 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C31 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| C32 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| A01 | 11FIC0401.MODE     | CAS  | 11S PURGE GAS FLOW                             | M      | M      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | 11FIC0401 IN CASCADE    |
| A02 | 11FIC0401.MODE     | AUT  | 11S PURGE GAS FLOW                             | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | 11FIC0401 IN AUTO       |
| A03 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |
| A04 |                    |      |                                                | .      | .      | . | . | . | . | . | . | . | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  | .  |                         |

# DCS Internal Logic Calculation

6) Watchdog Update: Annunciator added if 180 seconds timer up.

If Watchdog timer is up (180seconds) a message will be sent to DCS annunciator on communication fails. APC will not be able to switched on, if alarm reset button is not acknowledged.

**PCLDPE APC CONTROLLER**

APC TURN OFF CONDITION :  
 1. INITIATOR PUMP TESTING/FAIL  
 2. INITIATOR PUMP SWAPPING  
 3. APC WEB PAGE FAILURES  
 4. SECTIONAL DEFOULING  
 5. AD-HOC TRANSITION

| MAIN CONTROLLER STATUS | DESCRIPTION                | ONREQ | ONSTS | MAIN APC OFF SWITCH | WRDTIME |
|------------------------|----------------------------|-------|-------|---------------------|---------|
| Nonlinear Controller   | PCLDPE Main APC Controller | ON    | ON    | ON                  | RRR SEC |

APC DCS 180S  
TM ALM RST

**DCS - APC MV / DV MONITORING**

| DCS PID TAG   | DCS TIME CONSTANT FILTER | DESCRIPTIVE FOR APC                                     | BASE LAYER STATUS | APC PID MODE | ANTI-WINDUP | LOOP STATUS | MV OFF SWITCH | CL          |
|---------------|--------------------------|---------------------------------------------------------|-------------------|--------------|-------------|-------------|---------------|-------------|
| APCTA.SV      | RR                       | MV - CTA Gas Flow - 11FIC0602                           | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRR         |
| APTEMP1.SV    | RR                       | MV - Peak Temperature 1                                 | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R °C   |
| APTEMP2.SV    | RR                       | MV - Peak Temperature 2                                 | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R °C   |
| APTEMP3.SV    | RR                       | MV - Peak Temperature 3                                 | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R °C   |
| APTEMP4.SV    | RR                       | MV - Peak Temperature 4                                 | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R °C   |
| APPURGE.SV    | RR                       | MV - Purge Gas Flow                                     | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R kg/h |
| 13PI0107B.SV  | RR                       | MV - Extruder Dieplate Pressure                         |                   |              |             |             |               | RRRR.R Barg |
| 12PIC0103.PV  | RR                       | DV - Reactor Pressure                                   |                   |              |             |             |               | RRRR.R Barg |
| 12TIC0102A.PV | RR                       | DV - Preheater 2 Outlet Temp                            |                   |              |             |             |               | RRRR.R °C   |
| 12TIC0401.PV  | RR                       | DV - Reactor Hot Water Temperature                      |                   |              |             |             |               | RRRR.R °C   |
| DTALLZONE     | RR                       | DV - Total Diff Temp All Reactor Zone                   |                   | V1 RRRR.R    | V2 RRRR.R   | V3 RRRR.R   | V4 RRRR.R     | RRRR.R °C   |
| 11TI0302.PV   | RR                       | DV - 11-K103 Suction Temperature                        |                   |              |             |             |               | RRRR.R °C   |
| CTARATIO      | RR                       | DV - CTA / Ethylene Feed Flow ( 11FIC0602 / 39FI0102B ) |                   |              |             |             |               | RRRR.R      |
| 13TI0116B.PV  | RR                       | DV - Main Extruder Die Plate Temperature                |                   |              |             |             |               | RRRR.R °C   |
| 13JI0101.PV   | RR                       | DV - Main Ext Torque                                    |                   |              |             |             |               | RRRR.R %    |

Hidden Reset Alarm Button

OVERRIDE  
PEAK does not FOLLOW

**DCS - APC CV MONITORING**

| GRADE | MFI GOERTFERT | LAST MFI LAB     |       | GLOSS - LINE 1 AVG | GLOSS - LINE 2 AVG | PRODSPEX |
|-------|---------------|------------------|-------|--------------------|--------------------|----------|
| CCCCC | RRR.RR        | CCCCCCCCCCCCCCCC | CCCCC | RRRR.RR            | RRRR.RR            | RR.RR    |

# DCS Internal Logic Calculation

## 7) APC switch off button at DCS

New DCS APC page allows for panel to switch off MV's or APC Main controller. Approximate 1-2 minute delay for the status to reflect at APC PCWS. However, status at DCS shall be immediate. Switch ON is allowed only through PCWS.

### PCLDPE APC CONTROLLER

| MAIN CONTROLLER STATUS | DESCRIPTION                | ONREQ | ONSTS | MAIN APC OFF SWITCH | WRDTIME |
|------------------------|----------------------------|-------|-------|---------------------|---------|
| Nonlinear Controller   | PCLDPE Main APC Controller | ON    | ON    | ON                  | RRR SEC |

APC DCS 180S  
TM ALM RST

APC TURN OFF CONDITION :

1. INITIATOR PUMP TESTING/FAIL
2. INITIATOR PUMP SWAPPING
3. APC WEB PAGE FAILURES
4. SECTIONAL DESFOULING
5. AD-HOC TRANSITION

#### DCS - APC MV / DV MONITORING

| DCS PID TAG   | DCS TIME CONSTANT FILTER | DESCRIPTIVE FOR APC                                     | BASE LAYER STATUS | APC PID MODE | ANTI-WINDUP | LOOP STATUS | MV OFF SWITCH | CURRENT VALUE |
|---------------|--------------------------|---------------------------------------------------------|-------------------|--------------|-------------|-------------|---------------|---------------|
| APCTA.SV      | RR                       | MV - CTA Gas Flow - 11FIC0602                           | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R kg/h   |
| APTEMP1.SV    | RR                       | MV - Peak Temperature 1                                 | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R °C     |
| APTEMP2.SV    | RR                       | MV - Peak Temperature 2                                 | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R °C     |
| APTEMP3.SV    | RR                       | MV - Peak Temperature 3                                 | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R °C     |
| APTEMP4.SV    | RR                       | MV - Peak Temperature 4                                 | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R °C     |
| APPURGE.SV    | RR                       | MV - Purge Gas Flow                                     | AUTO              | IMAN         | HIGH        | DCS         | ON            | RRRR.R kg/h   |
| 13PI0107B.SV  | RR                       | MV - Extruder Dieplate Pressure                         |                   |              |             |             |               | RRRR.R Barg   |
| 12PIC0103.PV  | RR                       | DV - Reactor Pressure                                   |                   |              |             |             |               | RRRR.R Barg   |
| 12TIC0102A.PV | RR                       | DV - Preheater 2 Outlet Temp                            |                   |              |             |             |               | RRRR.R °C     |
| 12TIC0401.PV  | RR                       | DV - Reactor Hot Water Temperature                      |                   |              |             |             |               | RRRR.R °C     |
| DTALLZONE     | RR                       | DV - Total Diff Temp All Reactor Zone                   |                   | V1 RRRR.R    | V2 RRRR.R   | V3 RRRR.R   | V4 RRRR.R     | RRRR.R °C     |
| 11TI0302.PV   | RR                       | DV - 11-K103 Suction Temperature                        |                   |              |             |             |               | RRRR.R °C     |
| CTARATIO      | RR                       | DV - CTA / Ethylene Feed Flow ( 11FIC0602 / 39FI0102B ) |                   |              |             |             |               | RRRR.R        |
| 13TI0116B.PV  | RR                       | DV - Main Extruder Die Plate Temperature                |                   |              |             |             |               | RRRR.R °C     |
| 13JI0101.PV   | RR                       | DV - Main Ext Torque                                    |                   |              |             |             |               | RRRR.R %      |

#### DCS - APC CV MONITORING

| GRADE | MFI GOERTFERT | LAST MFI LAB     |       | GLOSS - LINE 1 AVG | GLOSS - LINE 2 AVG | PRODSPEX |
|-------|---------------|------------------|-------|--------------------|--------------------|----------|
| CCCCC | RRR.RR        | CCCCCCCCCCCCCCCC | CCCCC | RRRR.RR            | RRRR.RR            | RR.RR    |

New Switch Off Button

# DCS Internal Logic Calculation

### 8) Base Layer Status (MAN) during APC MV is On (ST3)

A new sequence table for Loop Status Timer is created for all MV's whereby, if MV Loop Status is in APC, however Base layer is in Manual for more than 420 seconds, DCS will send off request to update the loop status. This is to cater for response test activity.

| Sequence Tables (APTEMP3_ST3) |                    |      | Periodic Execution Type(T)<br>Output Only When Conditions Change(C)<br>Basic Scan<br>1<br>Control Phase<br>0 |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  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    |     |     |     |     |     |     |     |     |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |   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| No.                           | Tag name,Data item | Data | RuleNo.<br>StepNo.<br>Reference Signal Comment                                                               | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 | 181 | 182 | 183 | 184 | 185 | 186 | 187 | 188 | 189 | 190 | 191 | 192 | 193 | 194 | 195 | 196 | 197 | 198 | 199 | 200 | 201 | 202 | 203 | 204 | 205 | 206 | 207 | 208 | 209 | 210 | 211 | 212 | 213 | 214 | 215 | 216 | 217 | 218 | 219 | 220 | 221 | 222 | 223 | 224 | 225 | 226 | 227 | 228 | 229 | 230 | 231 | 232 | 233 | 234 | 235 | 236 | 237 | 238 | 239 | 240 | 241 | 242 | 243 | 244 | 245 | 246 | 247 | 248 | 249 | 250 | 251 | 252 | 253 | 254 | 255 | 256 | 257 | 258 | 259 | 260 | 261 | 262 | 263 | 264 | 265 | 266 | 267 | 268 | 269 | 270 | 271 | 272 | 273 | 274 | 275 | 276 | 277 | 278 | 279 | 280 | 281 | 282 | 283 | 284 | 285 | 286 | 287 | 288 | 289 | 290 | 291 | 292 | 293 | 294 | 295 | 296 | 297 | 298 | 299 | 300 | 301 | 302 | 303 | 304 | 305 | 306 | 307 | 308 | 309 | 310 | 311 | 312 | 313 | 314 | 315 | 316 | 317 | 318 | 319 | 320 | 321 | 322 | 323 | 324 | 325 | 326 | 327 | 328 | 329 | 330 | 331 | 332 | 333 | 334 | 335 | 336 | 337 | 338 | 339 | 340 | 341 | 342 | 343 | 344 | 345 | 346 | 347 | 348 | 349 | 350 | 351 | 352 | 353 | 354 | 355 | 356 | 357 | 358 | 359 | 360 | 361 | 362 | 363 | 364 | 365 | 366 | 367 | 368 | 369 | 370 | 371 | 372 | 373 | 374 | 375 | 376 | 377 | 378 | 379 | 380 | 381 | 382 | 383 | 384 | 385 | 386 | 387 | 388 | 389 | 390 | 391 | 392 | 393 | 394 | 395 | 396 | 397 | 398 | 399 | 400 | 401 | 402 | 403 | 404 | 405 | 406 | 407 | 408 | 409 | 410 | 411 | 412 | 413 | 414 | 415 | 416 | 417 | 418 | 419 | 420 | 421 | 422 | 423 | 424 | 425 | 426 | 427 | 428 | 429 | 430 | 431 | 432 | 433 | 434 | 435 | 436 | 437 | 438 | 439 | 440 | 441 | 442 | 443 | 444 | 445 | 446 | 447 | 448 | 449 | 450 | 451 | 452 | 453 | 454 | 455 | 456 | 457 | 458 | 459 | 460 | 461 | 462 | 463 | 464 | 465 | 466 | 467 | 468 | 469 | 470 | 471 | 472 | 473 | 474 | 475 | 476 | 477 | 478 | 479 | 480 | 481 | 482 | 483 | 484 | 485 | 486 | 487 | 488 | 489 | 490 | 491 | 492 | 493 | 494 | 495 | 496 | 497 | 498 | 499 | 500 | 501 | 502 | 503 | 504 | 505 | 506 | 507 | 508 | 509 | 510 | 511 | 512 | 513 | 514 | 515 | 516 | 517 | 518 | 519 | 520 | 521 | 522 | 523 | 524 | 525 | 526 | 527 | 528 | 529 | 530 | 531 | 532 | 533 | 534 | 535 | 536 | 537 | 538 | 539 | 540 | 541 | 542 | 543 | 544 | 545 | 546 | 547 | 548 | 549 | 550 | 551 | 552 | 553 | 554 | 555 | 556 | 557 | 558 | 559 | 560 | 561 | 562 | 563 | 564 | 565 | 566 | 567 | 568 | 569 | 570 | 571 | 572 | 573 | 574 | 575 | 576 | 577 | 578 | 579 | 580 | 581 | 582 | 583 | 584 | 585 | 586 | 587 | 588 | 589 | 590 | 591 | 592 | 593 | 594 | 595 | 596 | 597 | 598 | 599 | 600 | 601 | 602 | 603 | 604 | 605 | 606 | 607 | 608 | 609 | 610 | 611 | 612 | 613 | 614 | 615 | 616 | 617 | 618 | 619 | 620 | 621 | 622 | 623 | 624 | 625 | 626 | 627 | 628 | 629 | 630 | 631 | 632 | 633 | 634 | 635 | 636 | 637 | 638 | 639 | 640 | 641 | 642 | 643 | 644 | 645 | 646 | 647 | 648 | 649 | 650 | 651 | 652 | 653 | 654 | 655 | 656 | 657 | 658 | 659 | 660 | 661 | 662 | 663 | 664 | 665 | 666 | 667 | 668 | 669 | 670 | 671 | 672 | 673 | 674 | 675 | 676 | 677 | 678 | 679 | 680 | 681 | 682 | 683 | 684 | 685 | 686 | 687 | 688 | 689 | 690 | 691 | 692 | 693 | 694 | 695 | 696 | 697 | 698 | 699 | 700 | 701 | 702 | 703 | 704 | 705 | 706 | 707 | 708 | 709 | 710 | 711 | 712 | 713 | 714 | 715 | 716 | 717 | 718 | 719 | 720 | 721 | 722 | 723 | 724 | 725 | 726 | 727 | 728 | 729 | 730 | 731 | 732 | 733 | 734 | 735 | 736 | 737 | 738 | 739 | 740 | 741 | 742 | 743 | 744 | 745 | 746 | 747 | 748 | 749 | 750 | 751 | 752 | 753 | 754 | 755 | 756 | 757 | 758 | 759 | 760 | 761 | 762 | 763 | 764 | 765 | 766 | 767 | 768 | 769 | 770 | 771 | 772 | 773 | 774 | 775 | 776 | 777 | 778 | 779 | 780 | 781 | 782 | 783 | 784 | 785 | 786 | 787 | 788 | 789 | 790 | 791 | 792 | 793 | 794 | 795 | 796 | 797 | 798 | 799 | 800 | 801 | 802 | 803 | 804 | 805 | 806 | 807 | 808 | 809 | 810 | 811 | 812 | 813 | 814 | 815 | 816 | 817 | 818 | 819 | 820 | 821 | 822 | 823 | 824 | 825 | 826 | 827 | 828 | 829 | 830 | 831 | 832 | 833 | 834 | 835 | 836 | 837 | 838 | 839 | 840 | 841 | 842 | 843 | 844 | 845 | 846 | 847 | 848 | 849 | 850 | 851 | 852 | 853 | 854 | 855 | 856 | 857 | 858 | 859 | 860 | 861 | 862 | 863 | 864 | 865 | 866 | 867 | 868 | 869 | 870 | 871 | 872 | 873 | 874 | 875 | 876 | 877 | 878 | 879 | 880 | 881 | 882 | 883 | 884 | 885 | 886 | 887 | 888 | 889 | 890 | 891 | 892 | 893 | 894 | 895 | 896 | 897 | 898 | 899 | 900 | 901 | 902 | 903 | 904 | 905 | 906 | 907 | 908 | 909 | 910 | 911 | 912 | 913 | 914 | 915 | 916 | 917 | 918 | 919 | 920 | 921 | 922 | 923 | 924 | 925 | 926 | 927 | 928 | 929 | 930 | 931 | 932 | 933 | 934 | 935 | 936 | 937 | 938 | 939 | 940 | 941 | 942 | 943 | 944 | 945 | 946 | 947 | 948 | 949 | 950 | 951 | 952 | 953 | 954 | 955 | 956 | 957 | 958 | 959 | 960 | 961 | 962 | 963 | 964 | 965 | 966 | 967 | 968 | 969 | 970 | 971 | 972 | 973 | 974 | 975 | 976 | 977 | 978 | 979 | 980 | 981 | 982 | 983 | 984 | 985 | 986 | 987 | 988 | 989 | 990 | 991 | 992 | 993 | 994 | 995 | 996 | 997 | 998 | 999 | 1000 | 1001 | 1002 | 1003 | 1004 | 1005 | 1006 | 1007 | 1008 | 1009 | 1010 | 1011 | 1012 | 1013 | 1014 | 1015 | 1016 | 1017 | 1018 | 1019 | 1020 | 1021 | 1022 | 1023 | 1024 | 1025 | 1026 | 1027 | 1028 | 1029 | 1030 | 1031 | 1032 | 1033 | 1034 | 1035 | 1036 | 1037 | 1038 | 1039 | 1040 | 1041 | 1042 | 1043 | 1044 | 1045 | 1046 | 1047 | 1048 | 1049 | 1050 | 1051 | 1052 | 1053 | 1054 | 1055 | 1056 | 1057 | 1058 | 1059 | 1060 | 1061 | 1062 | 1063 | 1064 | 1065 | 1066 | 1067 | 1068 | 1069 | 1070 | 1071 | 1072 | 1073 | 1074 | 1075 | 1076 | 1077 | 1078 | 1079 | 1080 | 1081 | 1082 | 1083 | 1084 | 1085 | 1086 | 1087 | 1088 | 1089 | 1090 | 1091 | 1092 | 1093 | 1094 | 1095 | 1096 | 1097 | 1098 | 1099 | 1100 | 1101 | 1102 | 1103 | 1104 | 1105 | 1106 | 1107 | 1108 | 1109 | 1110 | 1111 | 1112 | 1113 | 1114 | 1115 | 1116 | 1117 | 1118 | 1119 | 1120 | 1121 | 1122 | 1123 | 1124 | 1125 | 1126 | 1127 | 1128 | 1129 | 1130 | 1131 | 1132 | 1133 | 1134 | 1135 | 1136 | 1137 | 1138 | 1139 | 1140 | 1141 | 1142 | 1143 | 1144 | 1145 | 1146 | 1147 | 1148 | 1149 | 1150 | 1151 | 1152 | 1153 | 1154 | 1155 | 1156 | 1157 | 1158 | 1159 | 1160 | 1161 | 1162 | 1163 | 1164 | 1165 | 1166 | 1167 | 1168 | 1169 | 1170 | 1171 | 1172 | 1173 | 1174 | 1175 | 1176 | 1177 | 1178 | 1179 | 1180 | 1181 | 1182 | 1183 | 1184 | 1185 | 1186 | 1187 | 1188 | 1189 | 1190 | 1191 | 1192 | 1193 | 1194 | 1195 | 1196 | 1197 | 1198 | 1199 | 1200 | 1201 | 1202 | 1203 | 1204 | 1205 | 1206 | 1207 | 1208 | 1209 | 1210 | 1211 | 1212 | 1213 | 1214 | 1215 | 1216 | 1217 | 1218 | 1219 | 1220 | 1221 | 1222 | 1223 | 1224 | 1225 | 1226 | 1227 | 1228 | 1229 | 1230 | 1231 | 1232 | 1233 | 1234 | 1235 | 1236 | 1237 | 1238 | 1239 | 1240 | 1241 | 1242 | 1243 | 1244 | 1245 | 1246 | 1247 | 1248 | 1249 | 1250 | 1251 | 1252 | 1253 | 1254 | 1255 | 1256 | 1257 | 1258 | 1259 | 1260 | 1261 | 1262 | 1263 | 1264 | 1265 | 1266 | 1267 | 1268 | 1269 | 1270 | 1271 | 1272 | 1273 | 1274 | 1275 | 1276 | 1277 | 1278 | 1279 | 1280 | 1281 | 1282 | 1283 | 1284 | 1285 | 1286 | 1287 | 1288 | 1289 | 1290 | 1291 | 1292 | 1293 | 1294 | 1295 | 1296 | 1297 | 1298 | 1299 | 1300 | 1301 | 1302 | 1303 | 1304 | 1305 | 1306 | 1307 | 1308 | 1309 | 1310 | 1311 | 1312 | 1313 | 1314 | 1315 | 1316 | 1317 | 1318 | 1319 | 1320 | 1321 | 1322 | 1323 | 1324 | 1325 | 1326 | 1327 | 1328 | 1329 | 1330 | 1331 | 1332 | 1333 | 1334 | 1335 | 1336 | 1337 | 1338 | 1339 | 1340 | 1341 | 1342 | 1343 | 1344 | 1345 | 1346 | 1347 | 1348 | 1349 | 1350 | 1351 | 1352 | 1353 | 1354 | 1355 | 1356 | 1357 | 1358 | 1359 | 1360 | 1361 | 1362 | 1363 | 1364 | 1365 | 1366 | 1367 | 1368 | 1369 | 1370 | 1371 | 1372 | 1373 | 1374 | 1375 | 1376 | 1377 | 1378 | 1379 | 1380 | 1381 | 1382 | 1383 | 1384 | 1385 | 1386 | 1387 | 1388 | 1389 | 1390 | 1391 | 1392 | 1393 | 1394 | 1395 | 1396 | 1397 | 1398 | 1399 | 1400 | 1401 | 1402 | 1403 | 1404 | 1405 | 1406 | 1407 | 1408 | 1409 | 1410 | 1411 | 1412 | 1413 | 1414 | 1415 | 1416 | 1417 | 1418 | 1419 | 1420 | 1421 | 1422 | 1423 | 1424 | 1425 | 1426 | 1427 | 1428 | 1429 | 1430 | 1431 | 1432 | 1433 | 1434 | 1435 | 1436 | 1437 | 1438 | 1439 | 1440 | 1441 | 1442 | 1443 | 1444 | 1445 | 1446 | 1447 | 1448 | 1449 | 1450 | 1451 | 1452 | 1453 | 1454 | 1455 | 1456 | 1457 | 1458 | 1459 | 1460 | 1461 | 1462 | 1463 | 1464 | 1465 | 1466 | 1467 | 1468 | 1469 | 1470 | 1471 | 1472 | 1473 | 1474 | 1475 | 1476 | 1477 | 1478 | 1479 | 1480 | 1481 | 1482 | 1483 | 1484 | 1485 | 1486 | 1487 | 1488 |

# DCS Internal Logic Calculation

## 9) TIC0950PRD\_CL Block Fix – Pump F (APC Running)

This block is a calculation block which was used in initial DCS response test. Based on system check, the last time the block was used in 2003. This block causes for for sudden peak drop during running Pump F due to the MV value was last updated to 173. Tracking the block value to actual current SV prevents the peak temp dip during running Pump F with APC Online.

### 5.9 TIC0950PRD\_CL Calculation Block

This calculation block force TIC0950PRD.MV to track 12TIC0950.SV

```
LDPE\FCS0102\DR0042\TIC0950PRD_CL.edf

General-Purpose Calculation Formula (TIC0950PRD_CL

Line Number   General-Purpose Calculation Formula
001           ALIAS MV TIC0950PRD.MV
002           ALIAS SV 12TIC0950.SV
003
004           MV = SV
```

FCS0101 DR0042 – TIC0950PRD\_CL



# Inferential Quality

# Inferential Quality (IQ)

There are 3 inferential have been developed and used directly as a measurement to APC. They are MFI and Gloss. Another one inferential , Diepress is used as one of the inputs to to MFI.

Online Apps APC: PETLIN\_ASV2020 (APC-APOLLO): All Variables: Operations 4/28/2021 12:04:49 PM

Application: PETLIN\_ASV2020 On On 4/28/2021 12:04:00 PM MISO Move Plan Calculation (4/28/2021 11:15:00 AM Controller On)

Independents Filter: None

| Name          | Description                 | Critical | Combined Status | Service Request | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Current Move |
|---------------|-----------------------------|----------|-----------------|-----------------|-------------|--------------------|-------------|---------------------|--------------------|--------------|
| APCTA.SV      | PROPYLENE TOK102 FLOW       | No       | Normal          | On              | On          | 360.000            | 423.848     | 470.000             | 414.468            | -2.452       |
| APTEMP1.SV    | REACTOR PEAK TEMP 1         | No       | Hi Limit        | On              | On          | 293.000            | 295.000     | 295.000             | 295.000            | 0.000        |
| APTEMP2.SV    | REACTOR PEAK TEMP 2         | No       | Hi Limit        | On              | On          | 293.000            | 295.000     | 295.000             | 295.000            | 0.000        |
| APTEMP3.SV    | REACTOR PEAK TEMP 3         | No       | Hi Limit        | On              | On          | 293.000            | 294.932     | 295.000             | 295.000            | 0.034        |
| APTEMP4.SV    | REACTOR PEAK TEMP 4         | No       | Hi Limit        | On              | On          | 288.000            | 291.000     | 291.000             | 291.000            | 0.000        |
| DIEPRESS.SV*  | 13A249 DIEPLATE PRESSURE    | No       | Normal          | On              | On          | 54.000             | 55.255      | 56.500              | 55.218             | -0.005       |
| APPURGE.SV    | S118 PURGE GAS FLOW         | No       | Lo Limit        | On              | On          | 950.000            | 950.000     | 1200.000            | 950.000            | 0.000        |
| 12PIC0103.PV  | REACTOR PRESSURE            | No       | Normal          | On              |             |                    | 2549.911    |                     | 2549.911           |              |
| 12TIC0102APV  | PREHEATER 2 TEMP            | No       | Normal          | On              |             |                    | 170.140     |                     | 170.140            |              |
| 12TIC0401.PV  | R201 RHW SUPPLY TEMP        | No       | Normal          | On              |             |                    | 169.950     |                     | 169.950            |              |
| DTALLZONE     | PEAK VALLEY TOTAL TEMP DIFF | No       | Normal          | On              |             |                    | 264.793     |                     | 264.793            |              |
| 11TI0302.PV   | K103 SUCT TEMPERATURE       | No       | Normal          | On              |             |                    | 37.402      |                     | 37.402             |              |
| CTARATIO      | CTA/ETHYLENE                | No       | Normal          | On              |             |                    | 14.313      |                     | 14.313             |              |
| 11FIC0602.PV* | PROPYLENE TOK102 FLOW       | No       | Normal          | On              |             |                    | 423.666     |                     | 423.666            |              |
| EXTORQUE*     | MAIN EXTRU TORQUE           | No       | Normal          | On              |             |                    | 33.464      |                     | 33.464             |              |
| 13TI0116B.PV  | 13A249 DIEPLATE TEMP        | No       | Normal          | On              |             |                    | 246.223     |                     | 246.223            |              |
| PRODSPEX.DV*  | PROD SPEED AFTER EXTRUDR    | No       | Normal          | On              |             |                    | 29.320      |                     | 29.320             |              |

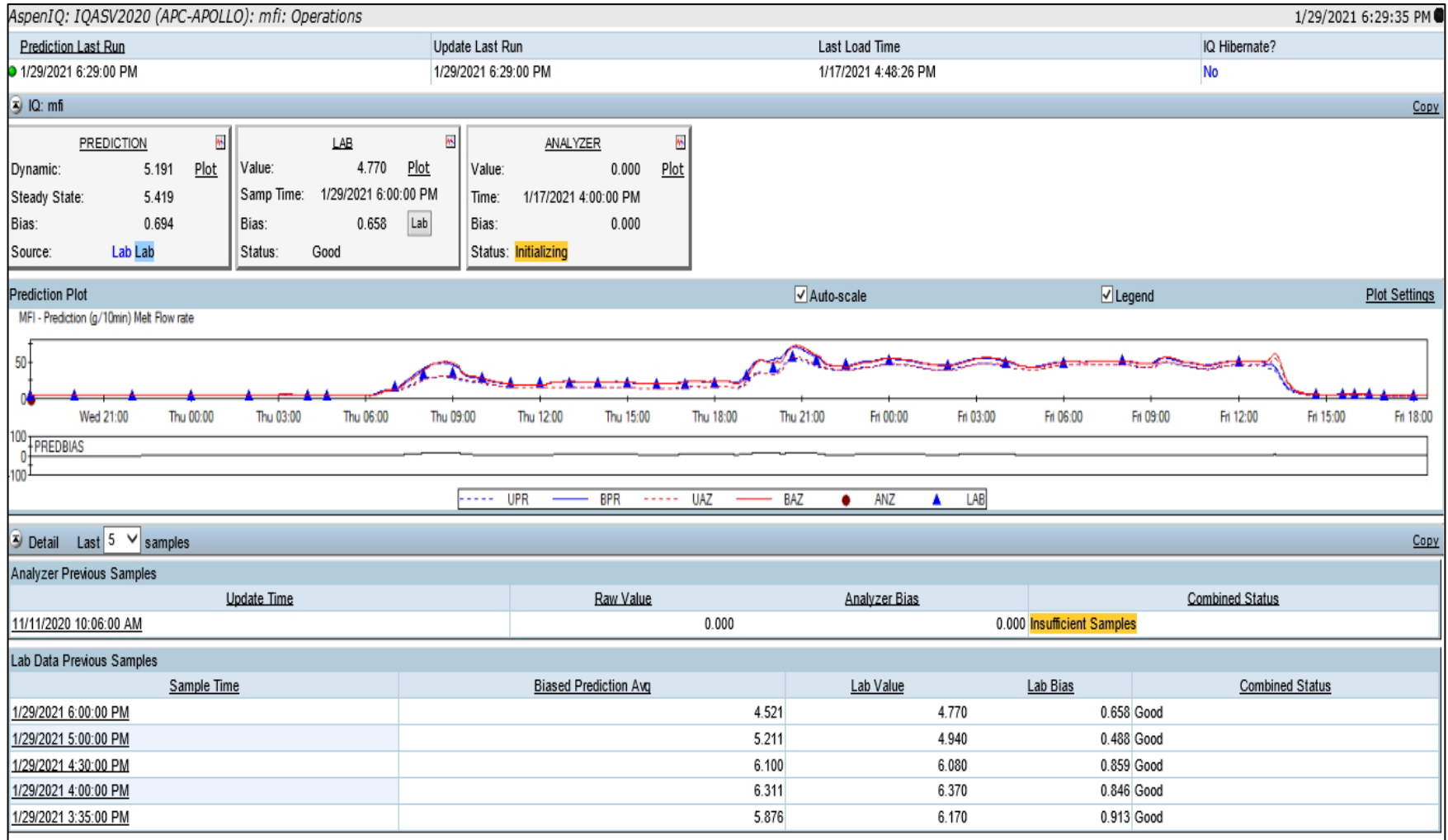
Dependents Filter: None

| Name        | Description                | Critical | Combined Status | Service Request | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Target |
|-------------|----------------------------|----------|-----------------|-----------------|--------------------|-------------|---------------------|--------------------|--------|
| PRODSPEX.PV | PROD SPEED AFTER EXTRUDR   | No       | < Lo Limit      | On              | 32.000             | 29.320      | 36.000              | 29.376             |        |
| MFI*        | Melt Flow rate             | No       | Ext Targ        | On              | 2.200              | 2.437       | 2.900               | 2.550              | 2.550  |
| DIEPRESS.PV | MAIN EXT DIEPLATE PRESSURE | No       | Ext Targ        | On              | 54.000             | 55.792      | 56.500              | 55.255             | 55.255 |
| GLOSS       | Gloss Line 1 - RUN         | No       | Normal          | On              | 52.000             | 59.961      | 70.000              | 60.189             |        |

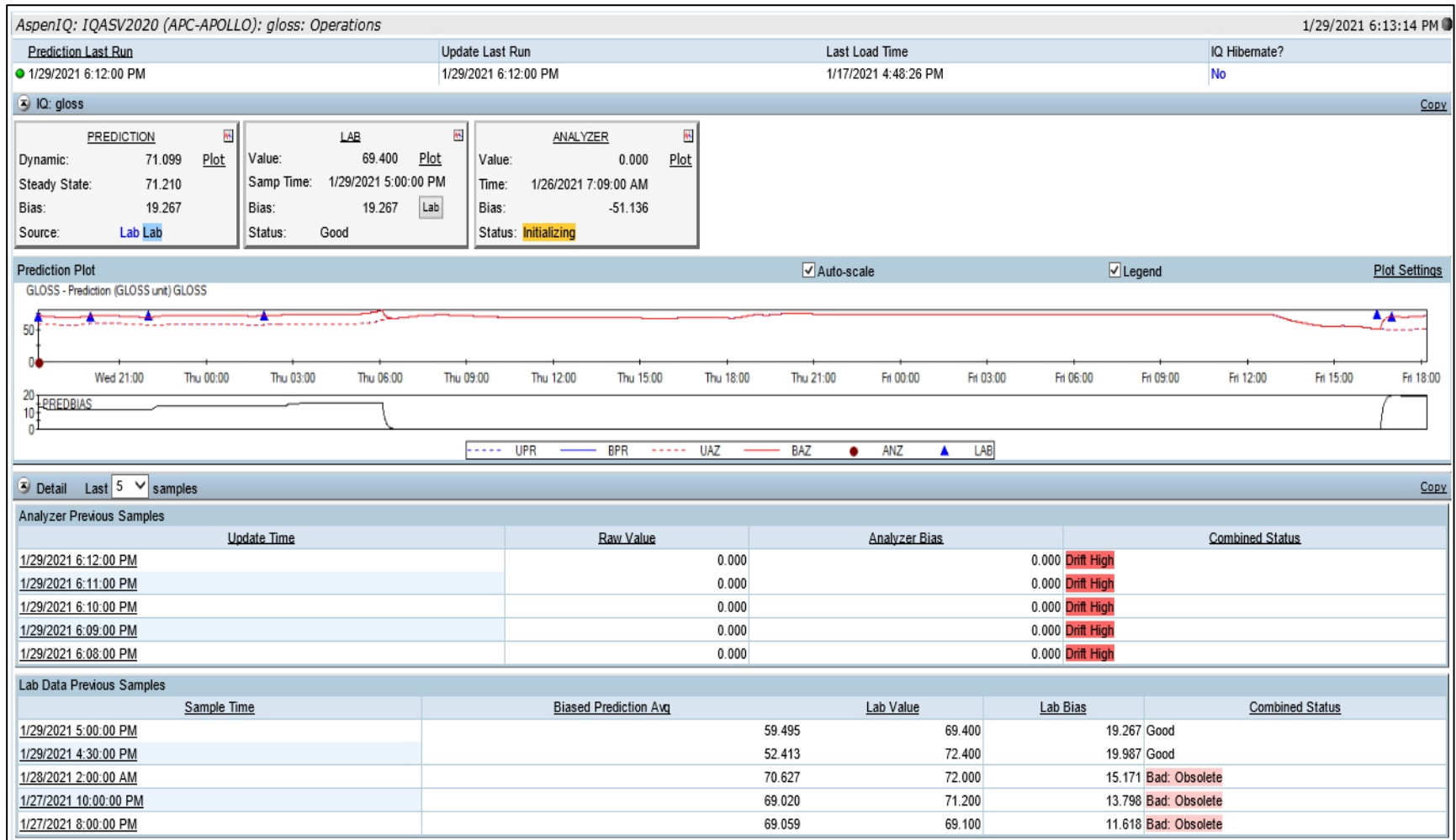
Messages Last 50

| Timestamp             | Application    | Message                                                                             |
|-----------------------|----------------|-------------------------------------------------------------------------------------|
| 4/28/2021 12:00:56 PM | PETLIN_ASV2020 | Entry at ind:APTEMP3.SV/ServiceRequest changed from Off to On.                      |
| 4/28/2021 12:00:56 PM | PETLIN_ASV2020 | User: Operator requested entry at ind:APTEMP3.SV/ServiceRequest be changed to: On.  |
| 4/28/2021 11:59:00 AM | PETLIN_ASV2020 | Entry at dep:DIEPRESS.PV/Target changed from 55.459 to 55.255.                      |
| 4/28/2021 11:38:22 AM | PETLIN_ASV2020 | Entry at ind:APTEMP3.SV/ServiceRequest changed from On to Off.                      |
| 4/28/2021 11:38:22 AM | PETLIN_ASV2020 | User: Operator requested entry at ind:APTEMP3.SV/ServiceRequest be changed to: Off. |

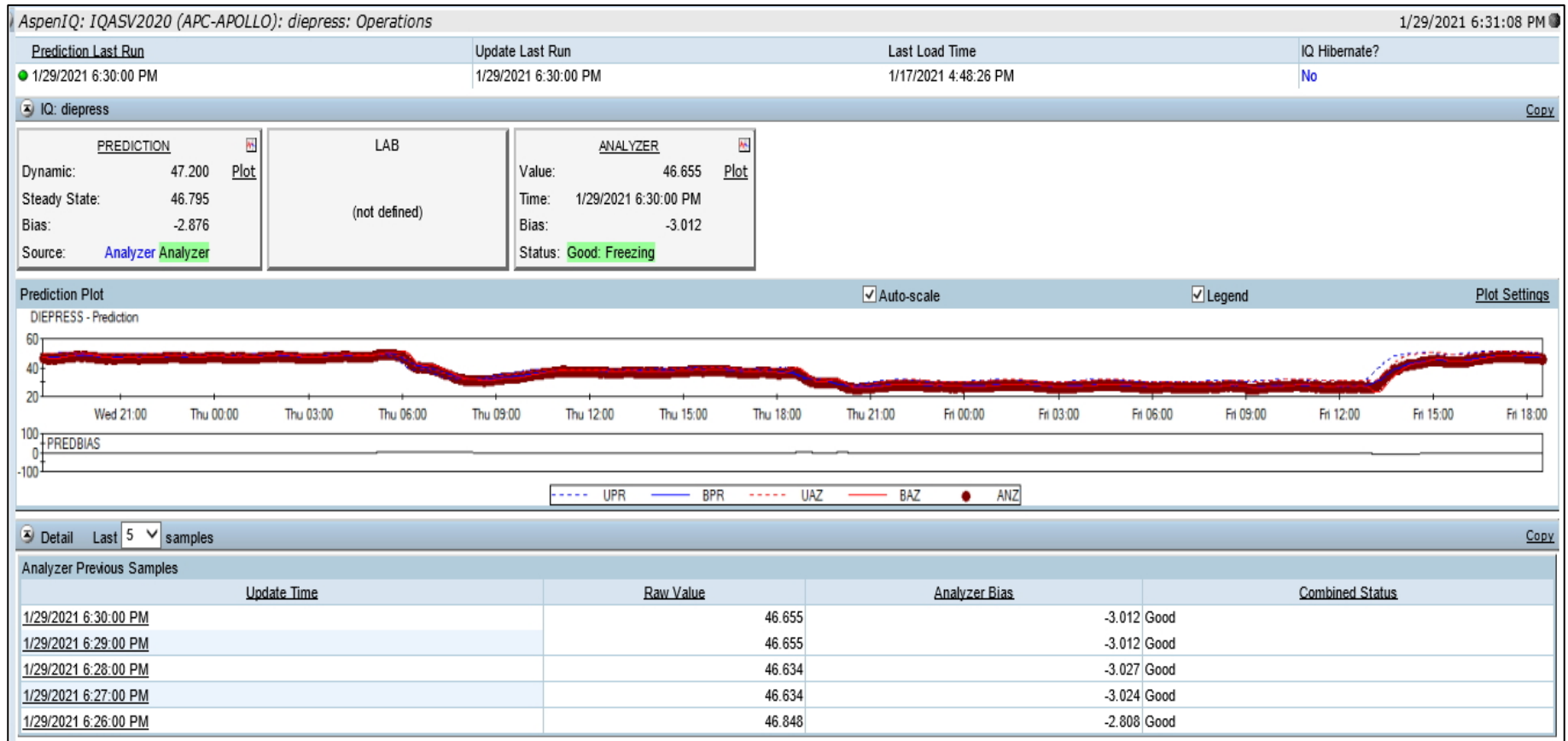
# MFI Inferential



# Gloss Inferential



# Diepress Inferential



# Inferential Quality (IQ)

## Lab Update

The following steps are crucial for APC to drive to correct solution:

- ☐ It is important to load APC QC lab result at least every 4 hours to update the inferential.
- ☐ Ensure that the lab data is correct before you update them into the model. Otherwise, it will give you bad prediction.
- ☐ Ensure that the APC QC lab data is collected at consistent time interval. Record the time of sample taken correctly & accurately!
- ☐ Ensure that you key in the correct time and lab value into Web Viewer. Please ensure there is no typo error!
- ☐ After the latest QC lab data is keyed in, you cannot key in the QC lab data which has already lapsed.
- ☐ If inferential update is not done, APC will not be able to correct the error (Bias Update) between the inferential prediction and actual value. This will give you bad prediction.
- ☐ You need to check the lab bias on web viewer to make sure that it is consistent.
- ☐ If 2 consecutive lab result sample is nearing product spec limit/off spec, directly to inform APC Focal for further assessment. Why?

# Inferential Quality (IQ)

## Analyzer Update

The following steps are crucial for APC to drive to correct solution:

- ☐ Operator to check analyzer status is good at Analyzer module in PCWS.
- ☐ Always counter check analyzer reading with lab result to ensure its accuracy.
- ☐ If the error between analyzer and lab reading is more than 5%, please request additional lab sample to confirm the reading.
- ☐ If it is consistent, please change the update source to lab.
- ☐ During preventive maintenance activity on the analyzer, please use lab reading as update source.

# How to update inferential lab measurement

Go to IQ ASV2020

1

2  
Select MFI

3 Select Lab tab and click "LAB" button

4  
Key in the time and Lab value

The screenshot displays the Aspen IQ software interface. The left sidebar shows a tree view with 'Aspen Process' expanded, leading to 'PETLIN\_ASV2020', then 'All Variables', 'Independents', 'Dependents', 'OPERATOR', 'Aspen IQ', 'IQASV2020', 'GLOSS', 'MFI', 'DIEPRESS', 'PRODSSEX', 'IQPETLIN', 'Aspen Process Sequencer', and 'HPTUBE'. The main area shows the 'LAB' tab selected, with a 'LAB' button highlighted. A dialog box titled 'Enter Lab Value' is open, showing a table of lab sample history and input fields for sample time and lab value.

| Sample Time           | Value | Pred  | Bias  | Status |
|-----------------------|-------|-------|-------|--------|
| 4/28/2021 10:10:00 AM | 2.710 | 3.199 | 0.320 | Good   |
| 4/28/2021 8:00:00 AM  | 2.090 | 2.391 | 0.470 | Good   |
| 4/28/2021 6:45:00 AM  | 2.060 | 1.913 | 0.595 | Good   |
| 4/28/2021 5:30:00 AM  | 2.380 | 4.035 | 0.689 | Good   |
| 4/28/2021 4:25:00 AM  | 1.790 | 0.964 | 0.769 | Good   |

The messages panel at the bottom shows several error messages related to 'Inferential MFI: Analyzer validation error, can't transform AVMINLIM.'

# How to update inferential lab measurement via Overview Tab

**1 Go to Overview tab**

**2 Click at the "LAB" button**

The screenshot displays the Aspen APC Web Interface. The 'Overview' tab is selected in the top navigation bar. The left sidebar shows a tree view of 'Online Apps' including 'Aspen Process Controller', 'Aspen IQ', and 'Aspen Process Sequencer'. The main content area contains several tables:

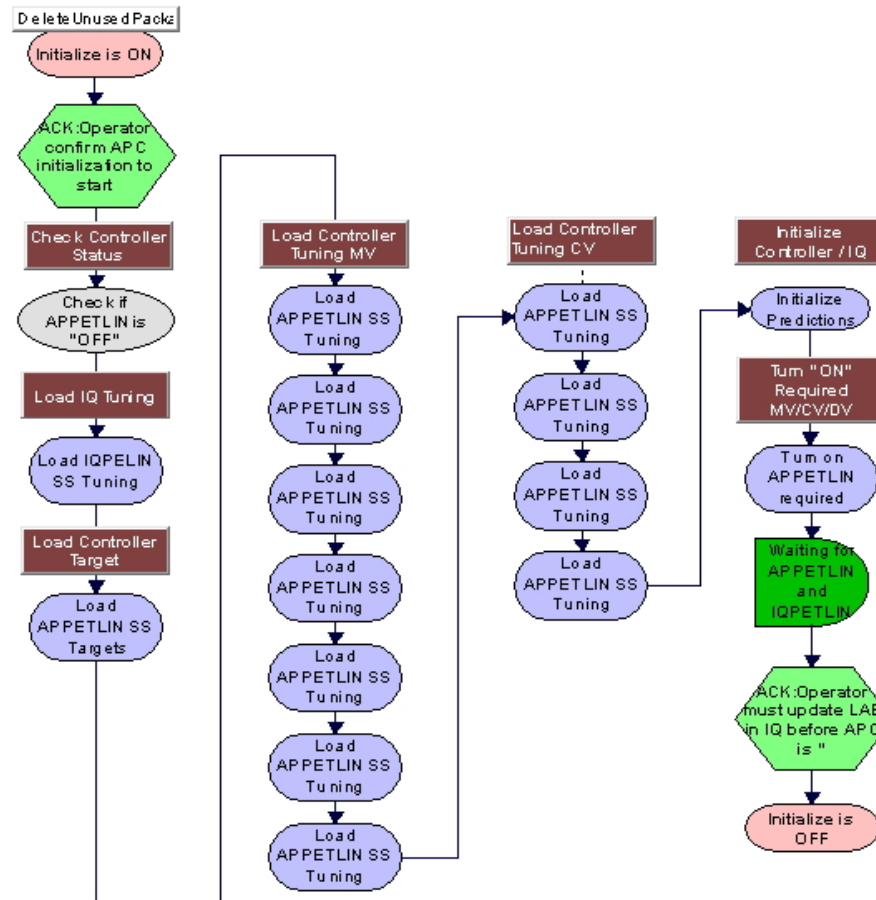
- Aspen Process Controller:** A table with columns: Name, Type, Master, Run State, Last Run Time, and Current Schedule. It lists 'PETLIN ASV2020' as an APC in a 'Running' state.
- Aspen IQ:** A table with columns: Name, Dyn Pred, SS Pred, Measurement, Bias Init, Last Run / Sample Time, and Update Source. It lists various inferential models like 'IQASV2020', 'GLOSS', 'MFI', 'DIEPRESS', 'PRODSPEX', and 'IQPETLIN'. Each model has a 'LAB' button next to its 'Update Source' column.
- Aspen Process Sequencer:** A table with columns: Equipment, Product, Run Plan, Strategy, Start Time, and End Time. It lists 'HPTUBE' as the equipment and 'C150Y' as the product.
- Messages:** A table with columns: Timestamp, Host, App Type, Name, and Message. It shows recent messages related to inferential model validation.

Annotations indicate the steps to update an inferential lab measurement: 1. Go to the Overview tab, and 2. Click at the "LAB" button in the Aspen IQ table.

# Grade Transition Manager

# Grade Transition Manager : *Grade Initialization*

Initialization:



# Grade Transition Manager :

## How to perform Grade Initialization

Aspentech Aspen APC Web Interface

Logoff | Help  
APC-APOLLO Administrator

Online History Optimizer Preferences Configuration

Overview | Faceplates | Operations | Engineering | Messages | Model | Calculations | Plots | Manage | Wizards

Online Apps APC: PETLIN\_ASV2020 (APC-APOLLO): All Variables: Operations 5/23/2021 4:02:18 PM

Application Last Run  
PETLIN\_ASV2020 On 5/23/2021 4:02:00 PM MISO Move Plan Calculation (5/19/2021 1:18:00 AM Controller On)

Independents Filter None Copy

| Name          | Description              | Critical | Combined Status | Service Request | Loop Status | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Current Move |
|---------------|--------------------------|----------|-----------------|-----------------|-------------|--------------------|-------------|---------------------|--------------------|--------------|
| APCTA.SV      | PROPYLENE TOK102 FLOW    | No       | Normal          | On              | On          | 480.000            | 531.528     | 570.000             | 529.029            | -0.308       |
| APTEMP1.SV    | REACTOR PEAK TEMP 1      | No       | Hi Limit        | On              | On          | 295.000            | 296.000     | 296.000             | 296.000            | 0.000        |
| APTEMP2.SV    | REACTOR PEAK TEMP 2      | No       | Hi Limit        | On              | On          | 295.000            | 296.000     | 296.000             | 296.000            | 0.000        |
| APTEMP3.SV    | REACTOR PEAK TEMP 3      | No       | Hi Limit        | On              | On          | 295.000            | 296.000     | 296.000             | 296.000            | 0.000        |
| APTEMP4.SV    | REACTOR PEAK TEMP 4      | No       | DCS Local       | Off             | Off         | 290.000            | 291.358     | 292.000             | 291.358            | 0.000        |
| DIEPRESS.SV*  | 13A249 DIEPLATE PRESSURE | No       | Hi Limit        | On              | On          | 46.500             | 47.967      | 48.000              | 48.000             | 0.001        |
| APPURGE.SV    | S118 PURGE GAS FLOW      | No       | Lo Limit        | On              | On          | 950.000            | 950.000     | 1100.000            | 950.000            | 0.000        |
| 12PIC0103.PV  | REACTOR PRESSURE         | No       | Normal          | On              |             |                    | 2499.952    |                     | 2499.952           |              |
| 12TIC0102APV  | PREHEATER 2 TEMP         | No       | Normal          | On              |             |                    | 167.078     |                     | 167.078            |              |
| 12TIC0401.PV  | R201 BHW SUPPLY TEMP     | No       | Normal          | On              |             |                    | 159.976     |                     | 159.976            |              |
|               |                          |          | Normal          | On              |             |                    | 291.526     |                     | 291.526            |              |
|               |                          |          | Normal          | On              |             |                    | 33.922      |                     | 33.922             |              |
|               |                          |          | Normal          | On              |             |                    | 16.242      |                     | 16.242             |              |
| 11FIC0602.PV* | PROPYLENE TOK102 FLOW    | No       | Normal          | On              |             |                    | 532.617     |                     | 532.617            |              |
| EXTORQUE*     | MAIN EXTRU TORQUE        | No       | Normal          | On              |             |                    | 27.810      |                     | 27.810             |              |
| 13TI0116B.PV  | 13A249 DIEPLATE TEMP     | No       | Normal          | On              |             |                    | 244.076     |                     | 244.076            |              |
| PRODSPEX.DV*  | PROD SPEED AFTER EXTRUDR | No       | Normal          | On              |             |                    | 32.383      |                     | 32.383             |              |

Dependents Filter None Copy

| Name        | Description              | Critical | Combined Status | Service Request | Operator Low Limit | Measurement | Operator High Limit | Steady State Value | Target |
|-------------|--------------------------|----------|-----------------|-----------------|--------------------|-------------|---------------------|--------------------|--------|
| PRODSPEX.PV | PROD SPEED AFTER EXTRUDR | No       | Normal          | On              | 32.000             | 32.383      | 36.000              | 32.339             |        |

1 Click on HPTUBE

# Grade Transition Manager :

## *How to perform Grade Initialization*

Aspen APC Web Interface  
Aspen APC Web Interface

Logoff | Help  
APC-APOLLO/Administrator

Online | History | Optimizer | Preferences | Configuration

Overview | Schedule | Activity List | Manage

Online Apps Aspen Process Sequencer : HPTUBE : Manage

Equipment: HPTUBE

| Product | Run Plan         | Strategy     | Start Time          | End Time |
|---------|------------------|--------------|---------------------|----------|
| C150Y   | Initialize_C150Y | Initialize_1 | 05/23/2021 15:59:04 |          |

Click on Manage

Click to delete any existing run plan

## Grade Transition Manager : *How to perform Grade Initialization*

Aspen APC Web Interface

Logoff | Help

APC-APOLLO Administrator

Online | History | Optimizer | Preferences | Configuration

Overview | Schedule | Activity List | Manage

Online Apps

Aspen Process Sequencer : HPTUBE : Manage

Equipment: HPTUBE

No product

Schedule a Transition ...

Aspen Process Controller

PETLIN\_ASV2020

- All Variables
- Independents
- Dependents
- OPERATOR

Aspen IQ

IQASV2020

- GLOSS
- MFI
- DIEPRESS
- PRODSPEX

Aspen Process Sequencer

- HPTUBE

Schedule a Transition -- Webpage Dialog

Equipment: HPTUBE

Current product: C150Y

New product: C170X

Start time: 05/23/2021 16:03

Run plan: C150Y\_To\_C170X

Strategy: Tx\_Main\_1

Initialize

OK Cancel

# Grade Transition Manager : *Grade Initialization*

Product: C150Y

| Activity Description                               | Action                              | Start | End   | Detail |
|----------------------------------------------------|-------------------------------------|-------|-------|--------|
| 1. APC Initialization Started                      | <input checked="" type="checkbox"/> | 15:59 | 00:01 |        |
| 2. [ACKNOWLEDGE] To Start Initialization           | <input type="checkbox"/>            | 15:59 | 16:00 |        |
| 3. APC Check if APPETLIN is "OFF"                  | <input type="checkbox"/>            | 16:00 | 16:01 |        |
| 4. Load IQ Tuning                                  | <input type="checkbox"/>            | 16:01 | 16:02 |        |
| 5. Load Controller Target                          | <input type="checkbox"/>            | 16:02 | 16:03 |        |
| 6. Load Controller Tuning MV                       | <input type="checkbox"/>            |       |       |        |
| 7. Load Controller Tuning CV                       | <input type="checkbox"/>            |       |       |        |
| 8. Load Controller Tuning DV                       | <input type="checkbox"/>            |       |       |        |
| 9. Initialize Controller / IQ                      | <input type="checkbox"/>            | 16:06 | 16:07 |        |
| 10. Turn "ON" Required MV/CV/DV                    | <input type="checkbox"/>            | 16:07 | 16:08 |        |
| 11. Waiting for Initialization to complete         | <input type="checkbox"/>            | 16:37 | 17:07 |        |
| 12. [ACKNOWLEDGE] Update LAB b4 switch bias to Lab | <input type="checkbox"/>            | 16:38 | 16:39 |        |
| 13. Complete APC Initialization                    | <input type="checkbox"/>            | 16:39 | 16:40 |        |

Click on to acknowledge grade initialization

Click on to complete initialization

What condition requires grade initialization:  
After Manual Transition & After Plant Start-up

When to start initialize Grade:  
After swing to prime silo

# Grade Transition Manager : *Grade Initialization*

## Hands-on Exercise

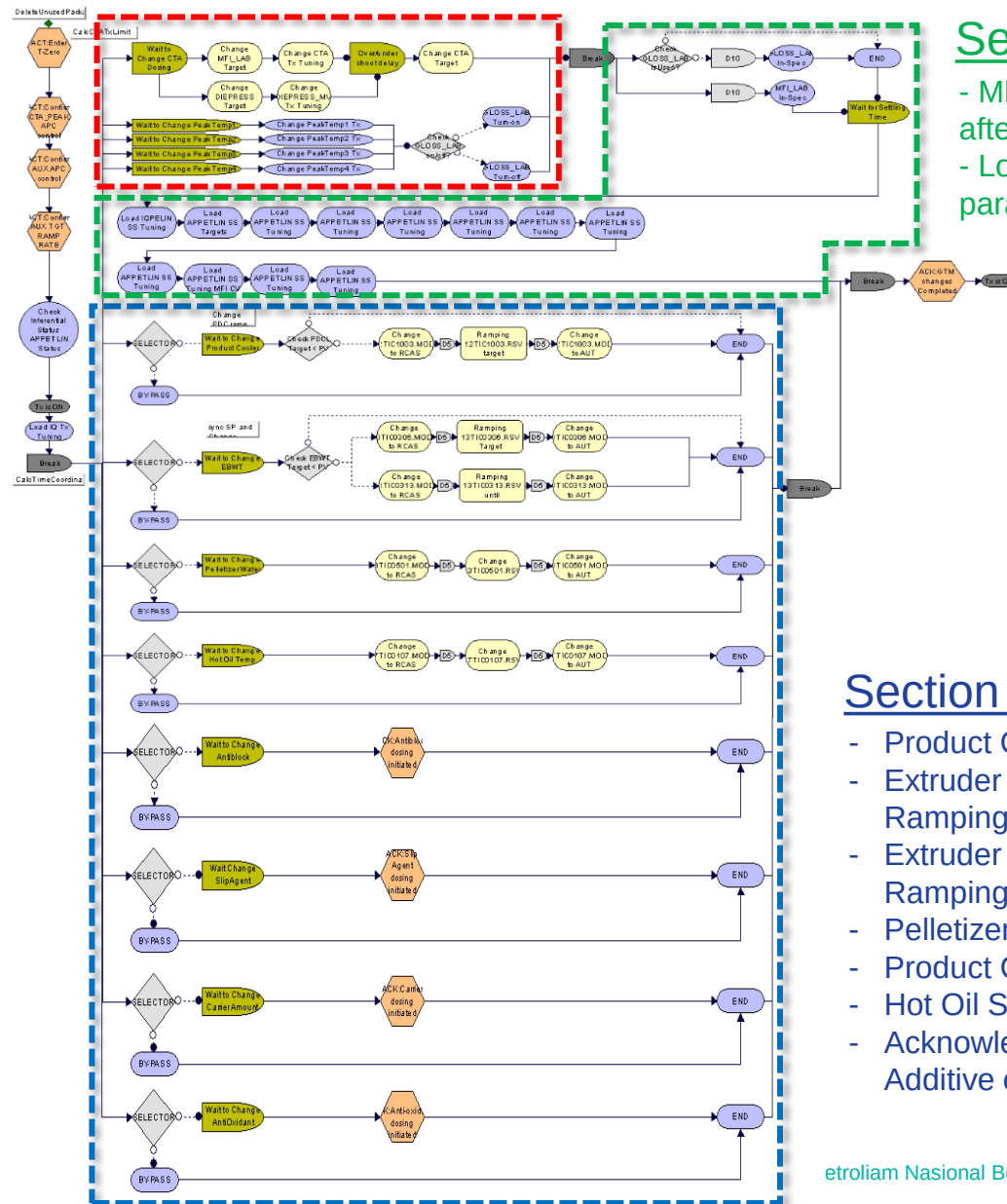
The following are actual exercise for actual grade initialization

- ☐ Hands-on Exercise 1:
  - ☐ Pre-requisite: APC System is already online
  - ☐ Activity: Start grade initialization (same grade to initialize is also allowed)
  - ☐ Queries: Can the initialization be completed?
  
- ☐ Hands-on Exercise 2:
  - ☐ Pre-requisite: APC System is OFFLINE (Turn APC Off temporarily if required.)
  - ☐ Activity: Start grade initialization (same grade to initialize is also allowed)
  - ☐ Queries:
    - ☐ Can the initialization be completed?
    - ☐ What parameter do you observed has been updated?

# Grade Transition Manager: Main Transition

## Section A

- CTA Overshoot -
- Peak Change -
- Prodspeex /Gloss On-Off -



## Section C

- MFI / Gloss Quality check after overshoot
- Load of APC steady state parameters (limits/tunings)

## Section B

- Product Cooler Ramping
- Extruder Barrel Water temp Ramping (13TIC0306)
- Extruder Barrel Heater Ramping (13TIC0313)
- Pelletizer Water
- Product Cooler SV change
- Hot Oil SV Change
- Acknowledgement button of Additive change

# GTM Activity List in PCWS

Aspen APC Web Interface  
Aspen APC Web Interface

Online | History | Optimizer | Preferences | Configuration

Overview | Schedule | Activity List | Manage

Online Apps

Aspen Process Controller

- PETLIN\_ASV2020
  - All Variables
  - Independents
  - Dependents
  - OPERATOR
- Aspen IQ
  - IQASV2020
    - GLOSS
    - MFI
    - DIEPRESS
    - PRODSPEX
  - IQPETLIN
- Aspen Process Sequencer
  - HPTUBE

Product: C150Y

**Section A**

**Section B**

**Section C**

| Activity Description                             | Action | Start | End   | Detail                                                                       |
|--------------------------------------------------|--------|-------|-------|------------------------------------------------------------------------------|
| 1. Enter DCS Transition Time                     |        | 11:16 | 00:01 |                                                                              |
| 2. Confirm CTA and PEAK APC control              |        | 11:16 | 11:18 | Wait to Change CTA Dosing<br>Time Remaining: 01:25:43<br>End Times: 13:38:01 |
| 3. Select AUXILIARY for APC Control              |        | 11:18 | 11:19 |                                                                              |
| 4. Confirm AUX target and ramp rate              |        | 11:18 | 11:19 | Wait to Change Peak Temp1<br>Time Remaining: 01:50:43<br>End Times: 14:03:01 |
| 5. APC Check Controller & Inferential Status     |        | 11:19 | 11:20 |                                                                              |
| ==SECTION A (CTA & PEAK CONTROL)==               |        | 00:00 | 00:01 |                                                                              |
| 6. Time to start CTA Over/Undershoot             |        | 00:00 | 00:01 | Wait to Change Peak Temp2<br>Time Remaining: 01:50:43<br>End Times: 14:03:01 |
| =6.1 CTA Over/undershoot & MFI Target Change     |        | 00:00 | 00:01 |                                                                              |
| =6.2 [COMPLETE] CTA over/undershoot              |        | 00:00 | 00:01 | Wait to Change Peak Temp3<br>Time Remaining: 01:50:43<br>End Times: 14:03:01 |
| 7. Time to Peak Temp1 Change                     |        | 00:00 | 00:01 |                                                                              |
| 8. Time to Peak Temp2 Change                     |        | 00:00 | 00:01 | Wait to Change Peak Temp4<br>Time Remaining: 01:50:43<br>End Times: 14:03:01 |
| 9. Time to Peak Temp3 Change                     |        | 00:00 | 00:01 |                                                                              |
| 10. Time to Peak Temp4 Change                    |        | 00:00 | 00:01 |                                                                              |
| ==SECTION B (AUXILIARY CONTROL)==                |        | 00:00 | 00:01 |                                                                              |
| 11. Time to Product Cooler Temp Change           |        | 00:00 | 00:01 |                                                                              |
| =11.1 [START] Ramping of 12TIC1003.SV            |        | 00:00 | 00:01 |                                                                              |
| =11.2 [COMPLETE] Ramping of 12TIC1003.SV         |        | 00:00 | 00:01 |                                                                              |
| 12. Time to Exit Barrel Water Temp Change        |        | 00:00 | 00:01 |                                                                              |
| =12.1 [START] Ramping of 13TIC0306.RSV           |        | 00:00 | 00:01 |                                                                              |
| =12.2 [COMPLETE] Ramping of 13TIC0306.RSV        |        | 00:00 | 00:01 |                                                                              |
| =12.3 [START] Ramping of 13TIC0313.RSV           |        | 00:00 | 00:01 |                                                                              |
| =12.4 [COMPLETE] Ramping of 13TIC0313.RSV        |        | 00:00 | 00:01 |                                                                              |
| 13. Time to Pelletizer Water Temp Change         |        | 00:00 | 00:01 |                                                                              |
| =13.1 [START] SV Change Pelletizer Water Temp    |        | 00:00 | 00:01 |                                                                              |
| =13.2 [COMPLETE] SV Change Pelletizer Water Temp |        | 00:00 | 00:01 |                                                                              |
| 14. Time to Hot Oil Temp Change                  |        | 00:00 | 00:01 |                                                                              |
| =14.1 [START] SV Change Hot Oil Temp             |        | 00:00 | 00:01 |                                                                              |
| =14.2 [COMPLETE] SV Change Hot Oil Temp          |        | 00:00 | 00:01 |                                                                              |
| 15. [ACKNOWLEDGE] Slip Agent dosing initiated    |        | 00:00 | 00:01 |                                                                              |
| 16. [ACKNOWLEDGE] Carrier dosing initiated       |        | 00:00 | 00:01 |                                                                              |
| 17. [ACKNOWLEDGE] Anti-block dosing initiated    |        | 00:00 | 00:01 |                                                                              |
| 18. [ACKNOWLEDGE] Anti-oxidant dosing initiated  |        | 00:00 | 00:01 |                                                                              |
| ==SECTION C (Quality, check and SS tuning)==     |        | 00:00 | 00:01 |                                                                              |
| 19. Settling time before quality Check           |        | 00:00 | 00:01 |                                                                              |
| 20. APC Check MFI Based Prediction In-Spec       |        | 00:00 | 00:01 |                                                                              |
| 21. APC Check GLOSS Based Prediction In-Spec     |        | 00:00 | 00:01 |                                                                              |
| 22. Wait for settling Time                       |        | 00:00 | 00:01 |                                                                              |
| 23. Load of New Grade APC SS Tuning and Limit    |        | 00:00 | 00:01 |                                                                              |
| ==COMPLETION OF ALL SECTION STATUS==             |        | 00:00 | 00:01 |                                                                              |
| 24. Sect-A (CTA & Peak Adj. Completed)           |        | 00:00 | 00:01 |                                                                              |
| 25. Sect-C (Load of new grade tuning completed)  |        | 00:00 | 00:01 |                                                                              |
| 26. Sect-B (Aux Adj. & Acknowledge Completed)    |        | 00:00 | 00:01 |                                                                              |
| 27. [ACKNOWLEDGE] Transition by GTM APC end      |        | 00:00 | 00:01 |                                                                              |

ACK/GTM changes Completed  
TimedOut - Waiting for operator response

==COMPLETION OF ALL SECTION STATUS==

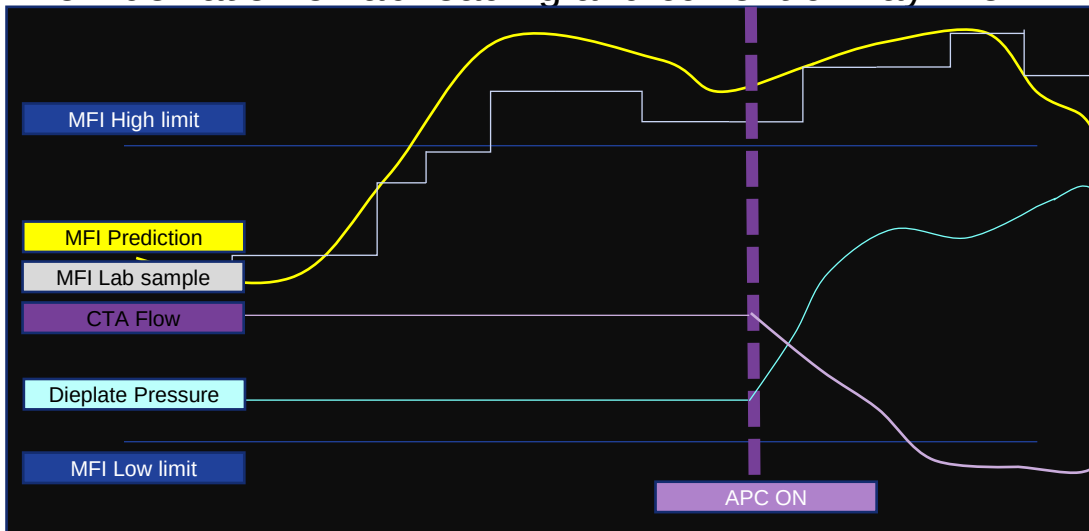
- 24. Sect-A (CTA & Peak Adj. Completed)
- 25. Sect-C (Load of new grade tuning completed)
- 26. Sect-B (Aux Adj. & Acknowledge Completed)
- 27. [ACKNOWLEDGE] Transition by GTM APC end

# PETRONAS

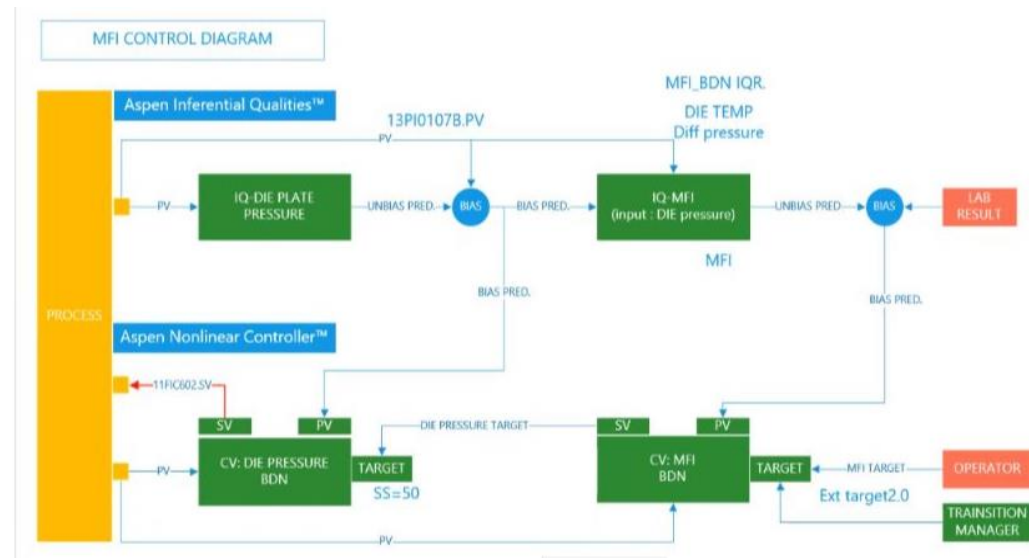
## Passionate about Progress

# Drawing

## How deviation of lab reading and current cv haywire



## The MFI control mechanism





# Aspen DMC3

Non-linear APC Modelling

Tips and Tricks Sharing Session

SEPT 2021

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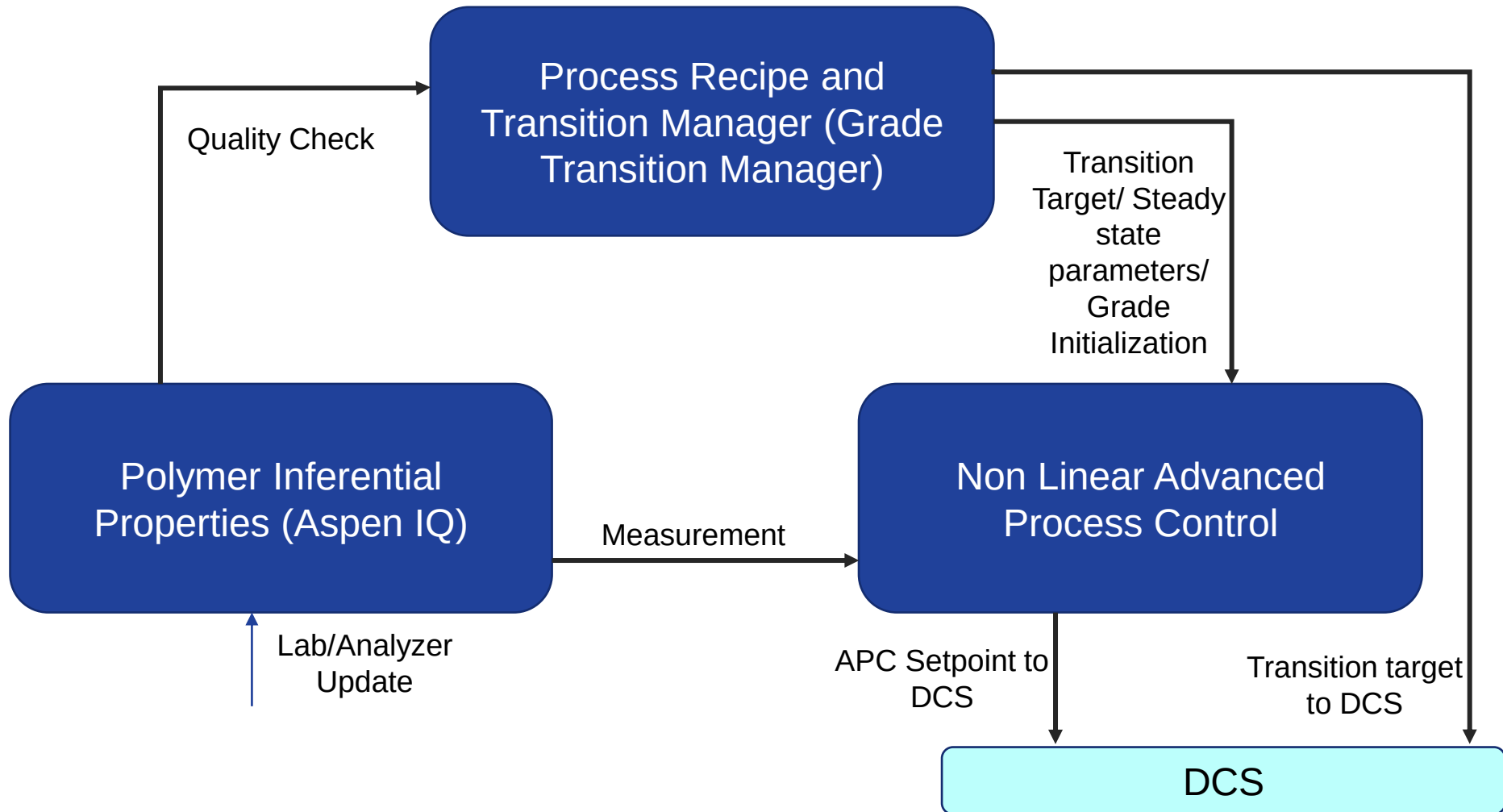
# Content

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# Objectives

- To discuss on PCLDPE APC Revamp development
- To understand the overview of nonlinear APC development
- To discuss best practice and lesson learnt
- Knowledge transfer and experience sharing within APC Fraternity

# PCLDPE APC Overview



# PCLDPE APC Overview

Advanced Process Control application was implemented in PCLDPE in order to achieve several objectives as below:

- ❑ Minimize variance in MFI and Gloss during steady state operation
- ❑ Minimize transition times by automating transitions
- ❑ Maximizing production rate without compromising quality
- ❑ Minimizing purge gas without compromising quality subject to constraints

# PCLDPE APC Overview

## Manipulated Variables

| No | Model Tag  | DCS Point   | Description      |
|----|------------|-------------|------------------|
| 1  | APCTA.SV   | APCTA.RSV   | Propylene Flow   |
| 2  | APTEMP1.SV | APTEMP1.RSV | Peak Temp Zone 1 |
| 3  | APTEMP2.SV | APTEMP2.RSV | Peak Temp Zone 2 |
| 4  | APTEMP3.SV | APTEMP3.RSV | Peak Temp Zone 3 |
| 5  | APTEMP4.SV | APTEMP4.RSV | Peak Temp Zone 4 |
| 6  | APPURGE.SV | APPURGE.RSV | Purge Gas Flow   |

## Controlled Variables

| No | Model Tag   | IP21 Point         | DCS Point         | Description        |
|----|-------------|--------------------|-------------------|--------------------|
| 1  | MFI         | IQ_MFI_BA          |                   | Melt Flow Index    |
| 2  | DIEPRESS.PV | IQ_DIEPRESS_BA .PV |                   | Die Plate Pressure |
| 3  | GLOSS       | IQ_GLOSS_BA        |                   | Gloss Lab          |
| 4  | PRODSPEX    |                    | PRODSPEX_PVFLT.PV | Production Rate    |

Yellow boxes are new variables included in APC Revamp 2020

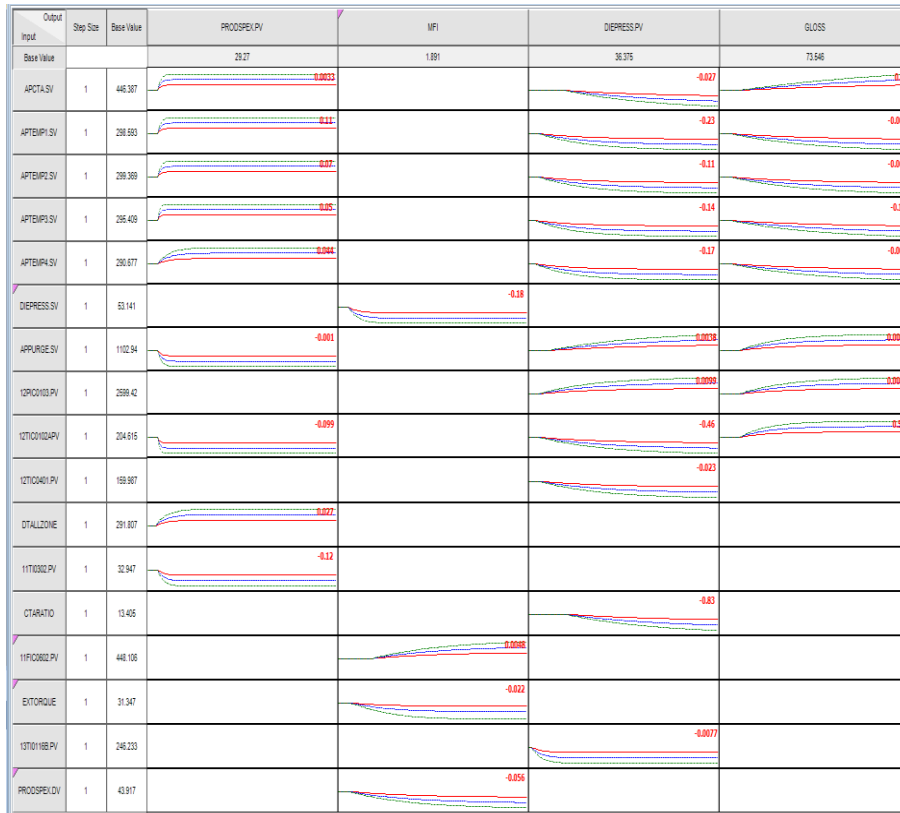
# PCLDPE APC Overview

Disturbance  
Variables

| No | Model Tag    | DCS Point           | Description                |
|----|--------------|---------------------|----------------------------|
| 1  | 12PIC0103SV  | 12PIC0103_PVFLT.PV  | Reactor Pressure           |
| 2  | 12TIC0401PV  | 12TIC0401_PVFLT.PV  | RHW Temp PV                |
| 3  | RXSTARTTEMP  | 12TIC0102A_PVFLT.PV | Rxn Start Temp (Preheater) |
| 4  | APTEMP1PV    | APTEMP1.RSV         | Peak Temp Zone 1           |
| 5  | APTEMP2PV    | APTEMP2.RSV         | Peak Temp Zone 2           |
| 6  | APTEMP3PV    | APTEMP3.RSV         | Peak Temp Zone 3           |
| 7  | APTEMP4PV    | APTEMP4.RSV         | Peak Temp Zone 4           |
| 8  | 11FIC0401.PV | APPURGE.RSV         | Purge Gas Flow             |
| 9  | CTARATIO     | CTARATIO_PVFLT.PV   | CTAFLOW/Ethylene Flow      |
| 10 | 13TI0116B.PV | 13TI0116B_PVFLT.PV  | Extruder Temperature       |
| 11 | 13JI0101.PV  | 13JI0101_PVFLT.PV   | Extruder Torque            |
| 12 | 11TI0302.PV  | 11TI0302_PVFLT.PV   | Suction temperature        |
| 13 | DTALLZONE    | DTALLZONE_PVFLT.PV  | Summation of all DIFFTEMP  |
| 14 | 11FIC0602.SV | 11FIC0602_PVFLT.PV  | CTA flow                   |
| 15 | PRODSPEX     | PRODSPEX_PVFLT.PV   | Production Rate            |

# PCLDPE APC Overview

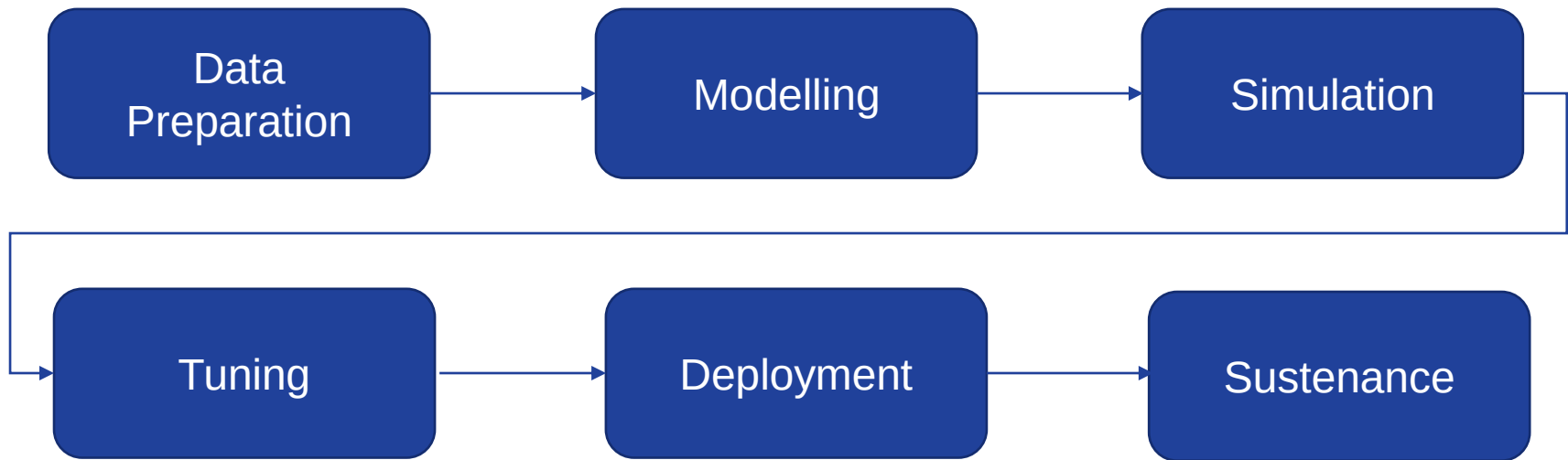
How APC know , what MVs to be moved and how much to be moved?



| CV | Variables     | PRODSPEX.PV | MFI | DIEPRESS.PV | GLOSS |
|----|---------------|-------------|-----|-------------|-------|
| MV | APCTA.SV      |             |     | -           |       |
|    | APTEMP1.SV    | +           |     |             | -     |
|    | APTEMP2.SV    | +           |     |             | -     |
|    | APTEMP3.SV    | +           |     |             | -     |
|    | APTEMP4.SV    | +           |     |             | -     |
|    | DIEPRESS.SV   |             | -   |             |       |
|    | APPURGE.SV    | -           |     |             | +     |
| DV | 11FIC0401.SV  |             |     | +           |       |
|    | 12PIC0103.PV  |             |     | +           | +     |
|    | 12TIC0102A.PV | -           |     | -           | +     |
|    | 12TIC0401.PV  |             |     | -           |       |
|    | APTEMP1.SV    |             |     | -           |       |
|    | APTEMP2.SV    |             |     | -           |       |
|    | APTEMP3.SV    |             |     | -           |       |
|    | APTEMP4.SV    |             |     | -           |       |
|    | DTALLZONE     | +           |     |             |       |
|    | 11TIO302.PV   | -           |     |             |       |
|    | CTARATIO      |             |     | -           |       |
|    | 11FIC0602.PV  | +           | +   |             | +     |
|    | 13JIO101.PV   |             | -   |             |       |
|    | 13TIO116B.PV  |             |     | -           |       |
|    | PRODSPEX.PV   |             | -   |             |       |

There is relationship between all MVs with the CVs ,and it is called APC model.

# Overall Workflow of Non-Linear APC



Generally, the workflow of developing Non-Linear APC is the same as linear controller.

The key different is modelling part. How it differ?

- ❖ Controller Preparation – Model ID, algo
- ❖ Inferential Qualities – How to build?

# Inferential Development

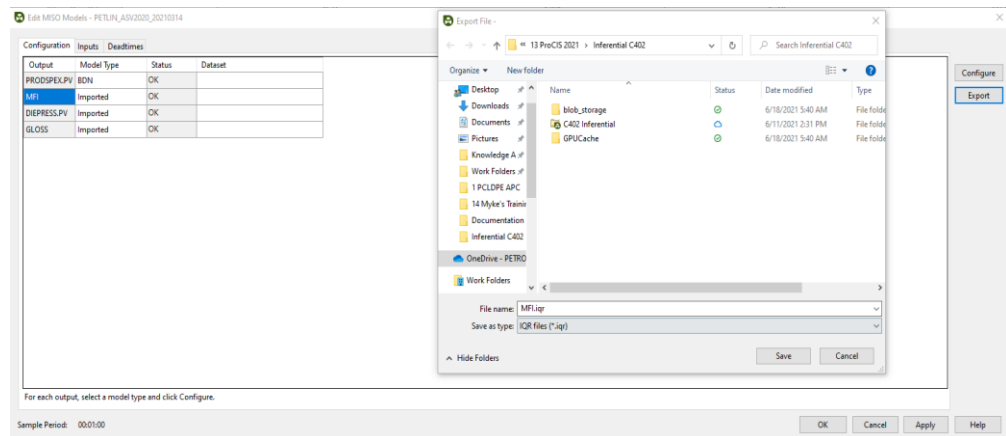
Inferential should be developed for qualities as measurement for APC.

Qualities:

1. Melt Flow Index (MFI)
2. Gloss
3. Diepress.SV

All qualities is developed in DMC3 Builder by using BDN. Then, the controllers file were exported as .iqr files (model file for inferential).

The .iqr file later will be used in .iqf (inferential configuration file) and uploaded as IQ variables.



What is the difference between typical way of building an inferential? - discuss

# Inferential Development

| Legacy (DMCPlus)                                                             | Non Linear Controller                                                                              | DMC3                                                                                                                  |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| DMCPlus is used to build the controller                                      | DMC non linear software is used to build to controller ( CVs)                                      | DMC3 linear software is used to build to controller ( CVs)                                                            |
| Different software is used to develop inferential – Aspen IQ                 | No need to develop inferential in Aspen IQ. Can use the same CV in controller by exporting IQ file | No need to develop inferential in Aspen IQ. Can use the same CV in controller by exporting IQ file/deep learning file |
| Configuration in DMCplus to link to Aspen IQ as measurement (CV measurement) | Configuration in DMC to link to Aspen IQ as measurement (CV measurement)                           | Configuration in DMC to link to Aspen IQ as measurement (CV measurement)                                              |

# The Basics – How / Where to Start?

## Data Preparation:

As per normal practice, historical data should be gathered for model identification. But how to get data for nonlinear process. Is it the same?

1. Historical vs Step Test data
  - a) Historical – For new APC, require a lot of data (1year-2year)
  - b) Step test – Revamp, clearer response and excitation, require more time to get the data
2. Step test at what grade?
  - a) To do step test for all grades.
3. Lab data vs Analyzer data.
  - a) Increase frequency of data sampling for lab data
  - b) Ensure good and reliable reading from analyzer
4. Data slicing for lab data? (slice more if the data look frozen)
  - a) To improve model accuracy and modeling process

# Beginning Modelling

Build new controller

Prediction Data - [APC] - Aspen DMC3 Builder - aspenONE

File Edit View Windows Help

Import New Controller Add Merge Step I/O Gain Recalculate Set Random Base Value Show Common Scale Generate Predictions Compare Models Edit Trans Variables

Search Aspen Knowledge

Controllers

PETLIN\_ASV2020 - Deploy PETLIN\_ASV2020\_20210314 - Models

Output Input Step Size Base Value PRODSPEX.PV

Base Value 29.27

APCTA.SV 1 448.387 -0.00 -0.0033

APTEMP1.SV 1 298.593 -0.6 -0.11

APTEMP2.SV 1 299.369 -0.6 -0.07

APTEMP3.SV 1 295.409 -0.6 -0.05

APTEMP4.SV 1 290.677 -0.6 -0.044

DIEPRESS.SV 1 53.141 -0.000 -0.001

APPURGE.SV 1 1102.94 -0.000 -0.001

12PIC0103.PV 1 2599.42 -0.0 -0.999

12TIC0102APV 1 204.615 -0.0 -0.999

12TIC0401.PV 1 159.987 -0.0 -0.999

DTALLZONE 1 291.807 -0.0 -0.027

11T10302.PV 1 32.947 -0.0 -0.12

CTARATIO 1 13.406 -0.0 -0.83

11FIC0602.PV 1 448.106 -0.00 -0.0048

EXTORQUE 1 31.347 -0.0 -0.022

13TIO116B.PV 1 245.233 -0.00 -0.0077

PRODSPEX.DV 1 43.917 -0.0 -0.056

Create Skeleton

Properties

Name: NewModel Model Type: Nonlinear

Sample Period: 1 Minutes Time to Steady State: 120 Minutes

Empty Model Select Variables from a Dataset

Variables

Dataset: master\_data\_full\_replicated

List: (All Vectors)

| Name         | Tag Name     | Input                    | Output                   | Units   | Description             |
|--------------|--------------|--------------------------|--------------------------|---------|-------------------------|
| 39FIO102B.PV | 39FIO102B.PV | <input type="checkbox"/> | <input type="checkbox"/> | T2ne/hr | MASS RATE               |
| 39TIO102.PV  | 39TIO102.PV  | <input type="checkbox"/> | <input type="checkbox"/> | degC    | ETHYLENE TEMPERATURE    |
| 11PI0303.PV  | 11PI0303.PV  | <input type="checkbox"/> | <input type="checkbox"/> | Barg    | K103 SUCTION PRESSURE   |
| 11FIC0602.SV | 11FIC0602.SV | <input type="checkbox"/> | <input type="checkbox"/> | kg/hr   | PROPYLENE TOK102 FLOW   |
| 11FIC0602.PV | 11FIC0602.PV | <input type="checkbox"/> | <input type="checkbox"/> | kg/hr   | PROPYLENE TOK102 FLOW   |
| 12TIC0720.SV | 12TIC0720.SV | <input type="checkbox"/> | <input type="checkbox"/> | degC    | P242A PEAK1 TEMP C2TROL |
| 12TIC0720.PV | 12TIC0720.PV | <input type="checkbox"/> | <input type="checkbox"/> | degC    | P242A PEAK1 TEMP C2TROL |
| 12TIC0730.SV | 12TIC0730.SV | <input type="checkbox"/> | <input type="checkbox"/> | degC    | P242B PEAK1 TEMP C2TROL |
| 12TIC0730.PV | 12TIC0730.PV | <input type="checkbox"/> | <input type="checkbox"/> | degC    | P242B PEAK1 TEMP C2TROL |
| 12TIC0820.SV | 12TIC0820.SV | <input type="checkbox"/> | <input type="checkbox"/> | degC    | P242C PEAK1 TEMP C2TROL |
| 12TIC0820.PV | 12TIC0820.PV | <input type="checkbox"/> | <input type="checkbox"/> | degC    | P242C PEAK1 TEMP C2TROL |

Key In time to steady state

Check the box for input or output

# Beginning Modelling

Click build model

Prediction Data - [APC] - Aspen DMC3 Builder - aspenONE

Search Aspen Knowledge

File Home Resource

Import New Controller Add Merge Controller Tree

Model Views Model Settings

Build Models Edit Model Gains

Generate Predictions Compare Models Edit Transforms Variables

Zoom In Zoom Out Zoom Full Width Plot Limits Plot Settings

Show Grid Lines Show Scale Labels Graphic Properties

master\_data\_full Set Random Time Show Value Line Dataset

Create Report

Controllers

PETLIN\_ASV2020

- Models
- Configuration
- Optimization
- Simulation
- Calculations
- Deployment

PETLIN\_ASV2020\_20210314

- Models
- Configuration
- Optimization
- Simulation
- Calculations
- Deployment

MFI

- Models
- Configuration
- Optimization
- Simulation
- Calculations
- Deployment

Output Input Step Size Base Value MFI

Base Value 0.00010 0.24

0.00010

Can check inputs and key in deadline

Can select different types of model type

Edit MISO Models - MFI

| Output | Model Type       | Status | Dataset                     |
|--------|------------------|--------|-----------------------------|
| MFI    | Linear Specified | OK     | master_data_full_replicated |

Configuration Inputs Deadtimes

For each output, select a model type and click Configure.

Sample Period: 00:01:00

PRODSPEX.PV 1 20.530 -0.00005 -0.00010

Configure Export

Configuration Inputs Deadtimes

| Output      | Model Type                    | Status | Dataset |
|-------------|-------------------------------|--------|---------|
| PRODSPEX.PV | BDN                           | OK     |         |
| MFI         | Linear Specified              | OK     |         |
| DIEPRESS.PV | BDN                           | OK     |         |
| GLOSS       | Generic Ratio Custom Imported | OK     |         |

OK Cancel Apply Help

# Beginning Modelling

Key in process data if required. We can manipulate based on our process understanding

Click "Configure"

Should get error and best error below than 0.02

For each output, select a model type and click Configure.

Sample Period: 00:01:00

Calibration Data ☒ Update Prediction Chart

Model Error ☒ Update Error Chart

Optimization Parameters

Max Iterations: 900 Random Seed: 1 Elements: 5

Gain Constraining Strategy

☒ Use steady state gain constraints only

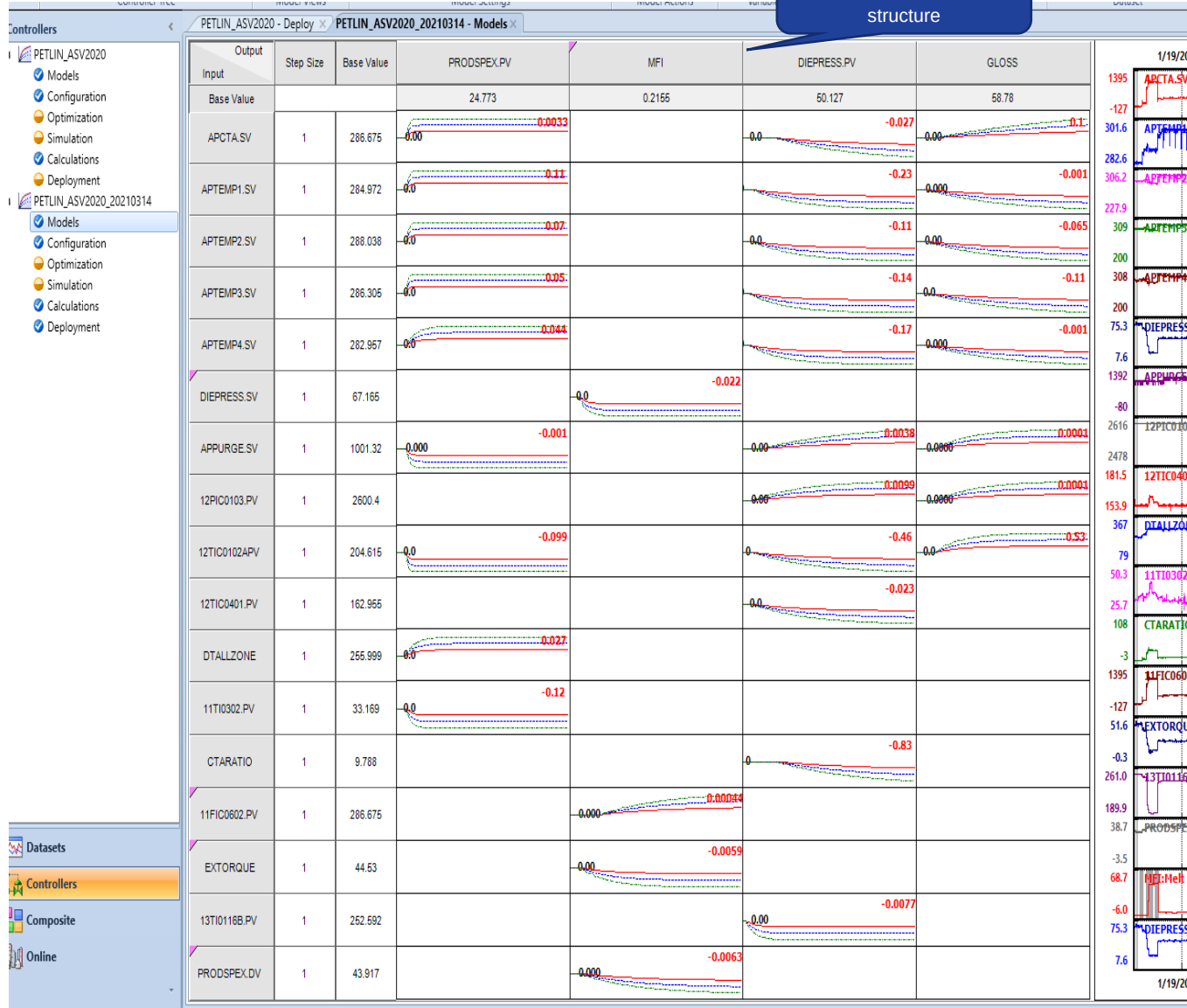
☐ Enforce monotonic states

Sample Period: 00:01:00

Close Help

# Beginning Modelling

Assemble all CVs, to  
have a full control  
structure



## Beginning Modelling – to assemble model

We can export our model.  
 .ady – dynamic file  
 .anl – steady state

## How to assemble model, in different file

The screenshot displays the Aspen Plus software interface. In the foreground, the 'Imported Model - MBI' dialog box is open, showing the 'Configuration' tab. This tab contains a table with columns for 'Output', 'Model Type', 'Status', and 'Dataset'. The table lists several components: PRODCSEXP.PV (Model Type: BEN, Status: OK), MBI (Model Type: Imported, Status: OK), DIEPRESS.PV (Model Type: OK, Status: OK), and GLOSS (Model Type: Imported, Status: OK). The 'Inputs' tab is also visible, showing a table with columns for 'Inputs', 'Model Type', 'Status', and 'Dataset'. The background shows the main Aspen Plus workspace with various process flow diagrams and data streams.

| Output       | Model Type | Status | Dataset |
|--------------|------------|--------|---------|
| PRODCSEXP.PV | BEN        | OK     |         |
| MBI          | Imported   | OK     |         |
| DIEPRESS.PV  |            | OK     |         |
| GLOSS        | Imported   | OK     |         |

# Simulation

Controller Tree      Optimization Views      Actions

Controllers <      PETLIN\_ASV2020 - Deploy      PETLIN\_ASV2020\_20210314 - Optimizer >

PETLIN\_ASV2020  
 Models  
 Configuration  
 Optimization  
 Simulation  
 Calculations  
 Deployment  
 PETLIN\_ASV2020\_20210314  
 Models  
 Configuration  
 Optimization  
 Simulation  
 Calculations  
 Deployment

Configure all optimization strategy first

| Input                    | Type | Target      | Description                 | Units   |
|--------------------------|------|-------------|-----------------------------|---------|
| APCTA.SV                 | MV   | None        | PROPYLENE TOK102 FLOW       | kg/hr   |
| APTEMP1.SV               | MV   | None        | REACTOR PEAK TEMP 1         | degC    |
| APTEMP2.SV               | MV   | None        | REACTOR PEAK TEMP 2         | degC    |
| APTEMP3.SV               | MV   | None        | REACTOR PEAK TEMP 3         | degC    |
| APTEMP4.SV               | MV   | None        | REACTOR PEAK TEMP 4         | degC    |
| DIEPRESS.SV              | MV   | None        | 13A249 DIEPLATE PRESSURE    | barg    |
| APPURGE.SV               | MV   | Lower Limit | S118 PURGE GAS FLOW         | kg/h    |
| 12PIC0103.PV             | DV   |             | REACTOR PRESSURE            | barg    |
| 12TIC0102APV             | DV   |             | PREHEATER 2 TEMP            | degC    |
| 12TIC0401.PV             | DV   |             | R201 RHW SUPPLY TEMP        | degC    |
| DTALLZONE                | DV   |             | PEAK VALLEY TOTAL TEMP DIFF | degC    |
| 11TI0302.PV              | DV   |             | K103 SUCT TEMPERATURE       | degC    |
| CTA/ETHYLENE             |      |             | CTA/ETHYLENE                |         |
| PROPYLENE TOK102 FLOW    |      |             | PROPYLENE TOK102 FLOW       | kg/hr   |
| MAIN EXTRU TORQUE        |      |             | MAIN EXTRU TORQUE           | %       |
| 13A249 DIEPLATE TEMP     |      |             | 13A249 DIEPLATE TEMP        | degC    |
| PROD SPEED AFTER EXTRUDR |      |             | PROD SPEED AFTER EXTRUDR    | tone/hr |

| Output      | Lower                               | Target          | Upper                               | Description                | Units      |
|-------------|-------------------------------------|-----------------|-------------------------------------|----------------------------|------------|
| PRODSPEX.PV | <input checked="" type="checkbox"/> | Upper Limit     | <input checked="" type="checkbox"/> | PROD SPEED AFTER EXTRUDR   | tone/hr    |
| MFI         | <input checked="" type="checkbox"/> | External Target | <input checked="" type="checkbox"/> | Melt Flow rate             | g/10min    |
| DIEPRESS.PV | <input checked="" type="checkbox"/> | External Target | <input checked="" type="checkbox"/> | MAIN EXT DIEPLATE PRESSURE | barg       |
| GLOSS       | <input checked="" type="checkbox"/> | None            | <input checked="" type="checkbox"/> | Gloss Line 1 - RUN         | gloss unit |

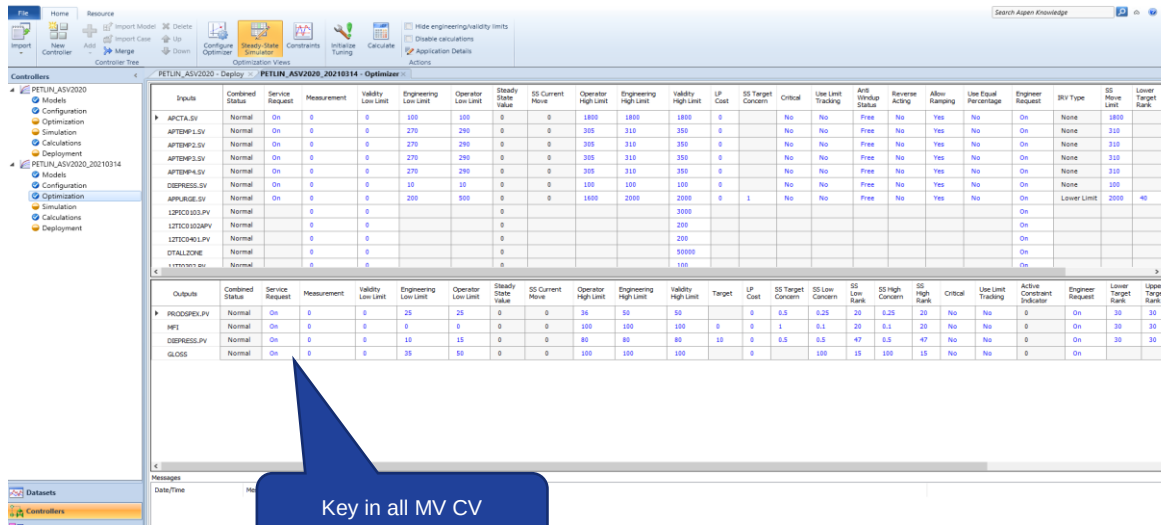
Datasets

Controllers

Composite

Online

# Simulation



Key in all MV CV parameters

| Inputs       | Combined Status | Service Request | Measurement | Validity Low Limit | Engineering Low Limit | Operator Low Limit | Steady State Value | SS Current Move | Operator High Limit | Engineering High Limit | Validity High Limit | LP Cost | SS Target Concern | SS Low Rank | SS High Rank | Critical | Use Limit Tracking | Active Constraint Indicator | Engineer Request | SS Type     | SS More Limit | Lower Target Rank |
|--------------|-----------------|-----------------|-------------|--------------------|-----------------------|--------------------|--------------------|-----------------|---------------------|------------------------|---------------------|---------|-------------------|-------------|--------------|----------|--------------------|-----------------------------|------------------|-------------|---------------|-------------------|
| APCTA.SV     | Normal          | On              | 0           | 0                  | 100                   | 100                | 0                  | 1800            | 1800                | 1800                   | 0                   | 0       | No                | No          | Free         | No       | Yes                | No                          | On               | None        | 310           |                   |
| APTEMP1.SV   | Normal          | On              | 0           | 0                  | 270                   | 290                | 0                  | 305             | 310                 | 350                    | 0                   | 0       | No                | No          | Free         | No       | Yes                | No                          | On               | None        | 310           |                   |
| APTEMP2.SV   | Normal          | On              | 0           | 0                  | 270                   | 290                | 0                  | 305             | 310                 | 350                    | 0                   | 0       | No                | No          | Free         | No       | Yes                | No                          | On               | None        | 310           |                   |
| APTEMP3.SV   | Normal          | On              | 0           | 0                  | 270                   | 290                | 0                  | 305             | 310                 | 350                    | 0                   | 0       | No                | No          | Free         | No       | Yes                | No                          | On               | None        | 310           |                   |
| APTEMP4.SV   | Normal          | On              | 0           | 0                  | 270                   | 290                | 0                  | 305             | 310                 | 350                    | 0                   | 0       | No                | No          | Free         | No       | Yes                | No                          | On               | None        | 310           |                   |
| DIAPRESS.SV  | Normal          | On              | 0           | 0                  | 10                    | 10                 | 0                  | 100             | 100                 | 100                    | 0                   | 1       | No                | No          | Free         | No       | Yes                | No                          | On               | None        | 100           |                   |
| DIAPRESS2.SV | Normal          | On              | 0           | 0                  | 200                   | 300                | 0                  | 1800            | 2000                | 2000                   | 0                   | 1       | No                | No          | Free         | No       | Yes                | No                          | On               | Lower Limit | 2000          | 40                |
| 12PIC103.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| DTALLZONE    | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12PIC105.PV  | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC103.MPV | Normal          | On              | 0           | 0                  | 0                     | 0                  | 0                  | 0               | 0                   | 0                      | 0                   | 0       | 0                 | 0           | 0            | 0        | 0                  | 0                           | On               | None        | 0             |                   |
| 12TIC104.LPV | Normal          | On              | 0           | 0                  | 0                     |                    |                    |                 |                     |                        |                     |         |                   |             |              |          |                    |                             |                  |             |               |                   |

# Tuning

PETLIN\_ASV2020 - Deploy > PETLIN\_ASV2020\_20210314 - Simulate (Malay Peninsula Standard Time)

Entry Type: ☒ Show All ☐ Inputs ☐ Results Value Type: ☐ Show All ☐ Operating Values ☒ Tuning Values

| Inputs       | Description               | Units   | Validity Low Limit | Engineering Low Limit | Engineering High Limit | Validity High Limit | LP Cost | SS Target Concern | Dynamic Target Concern | Move Down Limit | Move Up Limit | Reverse Acting | OutOfService Value | Model Bias | Allow Ramping | Use Equal Percentage | IRV Type    | SS Move Limit | Lower Target Rank | Upper Target Rank | High Eng Justification | Low Eng Justification | Horizon Length | Num Moves |
|--------------|---------------------------|---------|--------------------|-----------------------|------------------------|---------------------|---------|-------------------|------------------------|-----------------|---------------|----------------|--------------------|------------|---------------|----------------------|-------------|---------------|-------------------|-------------------|------------------------|-----------------------|----------------|-----------|
| APCTA.SV     | PROPYLENE TOK102 FLOW     | kg/hr   | 0                  | 100                   | 1800                   | 1800                | 0       | 1                 | 1                      | 1               | 1             | No             | 0                  | 0          | Yes           | No                   | None        | 1800          |                   |                   |                        |                       | 60             | 20        |
| APTEMP1.SV   | REACTOR PEAK TEMP 1       | degC    | 0                  | 270                   | 310                    | 350                 | 0       | 1                 | 1                      | 1               | 1             | No             | 0                  | 0          | Yes           | No                   | None        | 310           |                   |                   |                        |                       | 60             | 20        |
| APTEMP2.SV   | REACTOR PEAK TEMP 2       | degC    | 0                  | 270                   | 310                    | 350                 | 0       | 1                 | 1                      | 1               | 1             | No             | 0                  | 0          | Yes           | No                   | None        | 310           |                   |                   |                        |                       | 60             | 20        |
| APTEMP3.SV   | REACTOR PEAK TEMP 3       | degC    | 0                  | 270                   | 310                    | 350                 | 0       | 1                 | 1                      | 1               | 1             | No             | 0                  | 0          | Yes           | No                   | None        | 310           |                   |                   |                        |                       | 60             | 20        |
| APTEMP4.SV   | REACTOR PEAK TEMP 4       | degC    | 0                  | 270                   | 310                    | 350                 | 0       | 1                 | 1                      | 1               | 1             | No             | 0                  | 0          | Yes           | No                   | None        | 310           |                   |                   |                        |                       | 60             | 20        |
| DIEPRESS.SV  | 13A249 DIEPLATE           | barg    | 0                  | 10                    | 100                    | 100                 | 0       | 1                 | 1                      | 1               | 1             | No             | 0                  | 0          | Yes           | No                   | None        | 100           |                   |                   |                        |                       | 60             | 20        |
| APPURGE.SV   | S118 PURGE GAS            | kg/h    | 0                  | 200                   | 2000                   | 2000                | 0       | 1                 | 1                      | 1               | 1             | No             | 0                  | 0          | Yes           | No                   | Lower Limit | 2000          | 40                | 40                |                        |                       | 60             | 20        |
| 12PIC0103.PV | REACTOR PRESSURE          | barg    | 0                  |                       |                        | 3000                |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |
| 12TIC0102APV | PREHEATER                 |         |                    |                       |                        | 200                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |
| 12TIC0401.PV | R201 REACTOR              |         |                    |                       |                        | 200                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |
| DTALLZONE    | PEAK VIBRATION            |         |                    |                       |                        | 50000               |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |
| 11T0302.PV   | K103 SENSITIVITY          |         |                    |                       |                        | 100                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |
| CTARATIO     | CTA/ET                    |         |                    |                       |                        | 100                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |
| 11FIC0602.PV | PROPYLENE FLOW            |         |                    |                       |                        | 1800                |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |
| EXTORQUE     | MAIN EXTRU TORQUE         | %       | 0                  |                       |                        | 100                 |         |                   |                        |                 |               |                | 0                  | 0          |               |                      |             |               |                   |                   |                        |                       |                |           |
| 13T0116B.PV  | 13A249 DIEPLATE TEMP      | degC    | 0                  |                       |                        | 500                 |         |                   |                        |                 |               |                | 0                  | 0          |               |                      |             |               |                   |                   |                        |                       |                |           |
| PRODSPEX.DV  | PROD SPEED AFTER EXTRUDER | tone/hr | 0                  |                       |                        | 50                  |         |                   |                        |                 |               |                | 0                  | 0          |               |                      |             |               |                   |                   |                        |                       |                |           |

Tune the controller as per our controller intent

Typical setting that we normally tweak

PETLIN\_ASV2020 - Deploy > PETLIN\_ASV2020\_20210314 - Simulate (Malay Peninsula Standard Time)

Entry Type: ☒ Show All ☐ Inputs ☐ Results Value Type: ☐ Show All ☐ Operating Values ☒ Tuning Values

| Inputs       | Validity High Limit | LP Cost | SS Target Concern | Dynamic Target Concern | Move Down Limit | Move Up Limit | Reverse Acting | OutOfService Value | Model Bias | Allow Ramping | Use Equal Percentage | IRV Type    | SS Move Limit | Lower Target Rank | Upper Target Rank | High Eng Justification | Low Eng Justification | Horizon Length | Num Moves | Move Suppression | Move Suppression Increase | Move Resolution | Trajectory Factor | Transform Type         |
|--------------|---------------------|---------|-------------------|------------------------|-----------------|---------------|----------------|--------------------|------------|---------------|----------------------|-------------|---------------|-------------------|-------------------|------------------------|-----------------------|----------------|-----------|------------------|---------------------------|-----------------|-------------------|------------------------|
| APCTA.SV     | 1800                | 0       | 1                 | 1                      | 1               | No            | 0              | 0                  | 0          | Yes           | No                   | None        | 1800          |                   |                   |                        |                       | 60             | 20        | 1                | 2                         | 0               | 0                 |                        |
| APTEMP1.SV   | 150                 | 0       | 1                 | 1                      | 1               | No            | 0              | 0                  | 0          | Yes           | No                   | None        | 310           |                   |                   |                        |                       | 60             | 20        | 1                | 2                         | 0               | 0                 |                        |
| APTEMP2.SV   | 150                 | 0       | 1                 | 1                      | 1               | No            | 0              | 0                  | 0          | Yes           | No                   | None        | 310           |                   |                   |                        |                       | 60             | 20        | 1                | 2                         | 0               | 0                 |                        |
| APTEMP3.SV   | 150                 | 0       | 1                 | 1                      | 1               | No            | 0              | 0                  | 0          | Yes           | No                   | None        | 310           |                   |                   |                        |                       | 60             | 20        | 1                | 2                         | 0               | 0                 |                        |
| APTEMP4.SV   | 150                 | 0       | 1                 | 1                      | 1               | No            | 0              | 0                  | 0          | Yes           | No                   | None        | 310           |                   |                   |                        |                       | 60             | 20        | 1                | 2                         | 0               | 0                 |                        |
| DIEPRESS.SV  | 100                 | 0       | 1                 | 1                      | 1               | No            | 0              | 0                  | 0          | Yes           | No                   | None        | 100           |                   |                   |                        |                       | 60             | 20        | 1                | 2                         | 0               | 0                 | Modified Log Transform |
| APPURGE.SV   | 1000                | 0       | 1                 | 1                      | 1               | No            | 0              | 0                  | 0          | Yes           | No                   | Lower Limit | 2000          | 40                | 40                |                        |                       | 60             | 20        | 1                | 2                         | 0               | 0                 |                        |
| 12PIC0103.PV | 1000                |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |                  |                           |                 |                   |                        |
| 12TIC0102APV | 100                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |                  |                           |                 |                   |                        |
| 12TIC0401.PV | 100                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |                  |                           |                 |                   |                        |
| DTALLZONE    | 10000               |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |                  |                           |                 |                   |                        |
| 11T0302.PV   | 100                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |                  |                           |                 |                   |                        |
| CTARATIO     | 100                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |                  |                           |                 |                   |                        |
| 11FIC0602.PV | 800                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |                  |                           |                 |                   | Modified Log Transform |
| EXTORQUE     | 100                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |                  |                           |                 |                   | Modified Log Transform |
| 13T0116B.PV  | 100                 |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |                  |                           |                 |                   | Modified Log Transform |
| PRODSPEX.DV  | 10                  |         |                   |                        |                 |               |                |                    |            |               |                      |             |               |                   |                   |                        |                       |                |           |                  |                           |                 |                   | Modified Log Transform |

| Outputs     | SS High Rank | Dynamic Low Concern | Dynamic High Concern | Dynamic Target Concern | Deadtime Index | OutOfService Value | Time Constant Multiplier | Noise Ratio | Model Bias | Lower Target Rank | Upper Target Rank | Infeasible Limit Handling | Limit Straddle Handling | SS IRV Type     | Use Equal Percentage | High Eng Justification | Low Eng Justification | Horizon Length | Num Coincident Points | Coincident Point DeadZone | Trajectory Factor | Simulation Noise | Transform Type         |
|-------------|--------------|---------------------|----------------------|------------------------|----------------|--------------------|--------------------------|-------------|------------|-------------------|-------------------|---------------------------|-------------------------|-----------------|----------------------|------------------------|-----------------------|----------------|-----------------------|---------------------------|-------------------|------------------|------------------------|
| PRODSPEX.PV | 20           | 1                   | 1                    | 1                      | 0              | 0                  | 1                        | 0.1         | 0          | 30                | 30                | No                        | No                      | Upper Limit     | No                   |                        |                       | 120            | 20                    | 0                         | 0                 | 0                |                        |
| MFI         | 20           | 1                   | 1                    | 1                      | 0              | 0                  | 1                        | 0.1         | 0          | 30                | 30                | No                        | No                      | External Target | No                   |                        |                       | 120            | 20                    | 0                         | 0                 | 0                | Modified Log Transform |
| DIEPRESS.PV | 47           | 1                   | 1                    | 1                      | 0              | 0                  | 1                        | 0.1         | 0          | 30                | 30                | No                        | No                      | External Target | No                   |                        |                       | 120            | 20                    | 0                         | 0                 | 0                |                        |
| GLOSS       | 15           | 1                   | 1                    | 1                      | 0              | 0                  | 1                        | 0.1         | 0          |                   |                   | No                        | No                      | None            | No                   |                        |                       | 120            | 20                    | 0                         | 0                 | 0                |                        |

Detail Plots

Messages

# Deployment of controllers

To deploy the controllers from V11 to V8.7, we need to export

Deployment Overview in V11.

All CVs and MVs listed here.

Tags Configuration. Tags where APC read from.

Controllers

PETLIN\_ASV2020 - Deploy x PETLIN\_ASV2020\_20210314 - Deploy x

Tag Browser

Tag Generator

| Variable Name | Type    | Generate Tags                       | Measurement Prefix | Measurement Suffix | Interface Point |
|---------------|---------|-------------------------------------|--------------------|--------------------|-----------------|
| PETLIN_ASV... | General | <input checked="" type="checkbox"/> |                    |                    |                 |
| APCTA.SV      | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |
| APTEMP1.SV    | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |
| APTEMP2.SV    | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |
| APTEMP3.SV    | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |
| APTEMP4.SV    | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |
| DIEPRESS.SV   | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |
| APPURGE.SV    | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |
| 12PIC0103.PV  | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |
| 12TIC0102A... | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |
| 12TIC0401.PV  | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |
| DTALLZONE     | Input   | <input checked="" type="checkbox"/> |                    |                    |                 |

Variable Detail

| Parameter             | IO Source | IO Tag                            | IO Datatype | String Length | Test Value |
|-----------------------|-----------|-----------------------------------|-------------|---------------|------------|
| Master On Off Res...  | CIOEXAOPC | APOLLO_OR.PV                      | Integer     |               |            |
| Watchdog Count...     | CIOEXAOPC | APOLLO_WDRST.PV                   | Single      |               |            |
| Master On Off Sta...  | CIOEXAOPC | APOLLO_OS.PV                      | Integer     |               |            |
| Initialize Prediction | IOIP21    | AP_APPETLIN_INTRED IP_INPUT_VALUE | Integer     |               |            |
| TYA0930BPV            | CIOEXAOPC | 12TYA0930B.PV                     | Double      |               |            |
| TYA0950BPV            | CIOEXAOPC | 12TYA0950B.PV                     | Double      |               |            |
| 11FIC0602PVFLT        | CIOEXAOPC | 11FIC0602_PVFLT.PV                | Double      |               |            |
| 12TIC0720PVFLT        | CIOEXAOPC | APTEMP1_PVFLT.PV                  | Double      |               |            |
| 12TIC0820PVFLT        | CIOEXAOPC | APTEMP2_PVFLT.PV                  | Double      |               |            |
| 12TIC0920PVFLT        | CIOEXAOPC | APTEMP3_PVFLT.PV                  | Double      |               |            |
| 12TIC0950PVFLT        | CIOEXAOPC | APTEMP4_PVFLT.PV                  | Double      |               |            |
| 13PI0107BPVFLT        | CIOEXAOPC | 13PI0107B_PVFLT.PV                | Double      |               |            |

# Deployment on servers

Deployment Overview in V8.7.

Application loaded

Online controller

Aspen Process Controller Builder V8.7 - aspenONE

APC Project: PETLIN

Applications

- APPETLIN
- APPETLIN[20200227\_152157]
- APPETLIN[20201027\_162847]
- PETLIN
- PETLIN\_ASV2020
- PETLIN\_ASV2020[20201202\_122251]
- PETLIN\_ASV2020[20201209\_115335]
- PETLIN\_ASV2020[20201210\_142204]
- PETLIN\_ASV2020[20201231\_233106]
- PETLIN\_ASV2020[20210117\_150536]
- PETLIN\_ASV2020[20210308\_095000]
- PETLIN\_ASV2020[20210324\_171347]
- PETLIN\_ASV2020[20210517\_095224]
- PETLIN\_ASV2020[20210718\_151640]
- PETLIN\_ASV2020\_01172021
- PETLIN\_ASV2020\_20201202
- PETLIN\_TEST\_CALC[20201101\_092927]

Datasets

- MASTERDATASET

Models

Online Applications

| Name           | Type | Server     | Last Load Time        | Last Run Time         | Status  | Auto-Start                          |
|----------------|------|------------|-----------------------|-----------------------|---------|-------------------------------------|
| PETLIN_ASV2020 | APC  | APC-APOLLO | 5/18/2021 4:58:30 ... | 7/18/2021 3:17:00 ... | Running | <input checked="" type="checkbox"/> |

Start Stop Delete Get Snapshot... Get History...

Recent Operations

| Application    | Action   | Start Time      | Update Time          | Status    | % Complete | Server     |
|----------------|----------|-----------------|----------------------|-----------|------------|------------|
| PETLIN_ASV2020 | Snapshot | 7/18/2021 3:... | 7/18/2021 3:17:03 PM | Completed | 100        | APC-APOLLO |
| PETLIN_ASV2020 | Deploy   | 7/18/2021 3:... | 7/18/2021 3:13:20 PM | Cancelled | 0          |            |
| PETLIN_ASV2020 | Deploy   | 7/18/2021 3:... | 7/18/2021 3:12:22 PM | Cancelled | 0          |            |

Aspen Process Controller Builder V8.7 - aspenONE

APC Project: PETLIN

Applications

- APPETLIN
- APPETLIN[20200227\_152157]
- APPETLIN[20201027\_162847]
- PETLIN
- PETLIN\_ASV2020
- PETLIN\_ASV2020[20201202\_122251]
- PETLIN\_ASV2020[20201209\_115335]
- PETLIN\_ASV2020[20201210\_142204]
- PETLIN\_ASV2020[20201231\_233106]
- PETLIN\_ASV2020[20210117\_150536]
- PETLIN\_ASV2020[20210308\_095000]
- PETLIN\_ASV2020[20210324\_171347]
- PETLIN\_ASV2020[20210517\_095224]
- PETLIN\_ASV2020[20210718\_151640]
- PETLIN\_ASV2020\_01172021
- PETLIN\_ASV2020\_20201202
- PETLIN\_TEST\_CALC[20201101\_092927]

Datasets

- MASTERDATASET

Models

Online Applications

| Name           | Type | Server     | Last Load Time        | Last Run Time         | Status  | Auto-Start                          |
|----------------|------|------------|-----------------------|-----------------------|---------|-------------------------------------|
| PETLIN_ASV2020 | APC  | APC-APOLLO | 5/18/2021 4:58:30 ... | 7/18/2021 3:16:00 ... | Running | <input checked="" type="checkbox"/> |

Start Stop Delete Get Snapshot... Get History...

Get Application Snapshot

Selected Application Name: PETLIN\_ASV2020

Online Server: APC-APOLLO

Snapshots

☐ Select Available (Malay Peninsula Standard Time)

- 7/18/2021 3:01:00 PM
- 7/18/2021 11:01:00 AM
- 7/18/2021 7:00:00 AM
- 7/18/2021 3:00:00 AM
- 7/17/2021 10:59:00 PM
- 7/17/2021 6:59:00 PM

☒ Generate New

Destination Application

Save application snapshot using the following name:

PETLIN\_ASV2020[20210718\_151640]

OK Cancel Help

Update Time

| Update Time       | % Complete | Server     |
|-------------------|------------|------------|
| 8/2021 3:16:36 PM | 0          | APC-APOLLO |
| 8/2021 3:13:20 PM |            |            |
| 8/2021 3:12:22 PM |            |            |

Create backup by generating snapshot

# Deployment on servers

The screenshot displays the Aspen Process Controller Builder V8.7 - aspenONE interface. The left pane shows the project structure for 'APC Project: PETLIN', including Applications, Datasets, and Models. The main workspace is divided into several tabs: Recycle Bin, Identify, Extend, Optimize, Control, and Connect. The 'Control' tab is active, showing a table of controllers. A context menu is open over the 'PETLIN\_ASV2020' controller, listing options such as 'Configure Connection', 'Deploy', 'Edit Calculations', 'Save Tuning Set', 'Export Application', 'Save Application to Project', 'Create Report File', 'Show Application Differences', 'Copy', 'Delete', and 'Undo Edit Deploy Options'. A blue callout points to this menu with the text 'All settings for controller'.

Below the main interface, a 'Deploy - PETLIN\_ASV2020' dialog box is shown. It contains the following information:

- Application name: PETLIN\_ASV2020
- Server: APC-APOLLO (selected from a dropdown)
- Status: This server is available
- ☐ Enable online simulation
- ☒ Enable Aspen Watch monitoring
- Buttons: Redeploy, Cancel, Help

A blue callout points to the dialog box with the text 'Controller ready to be deployed'.

# Grade Transition Manager

Aspen APC Web Interface  
Aspen APC Web Interface

Logoff | Help  
APC-APOLLO Administrator

Online | History | Optimizer | Preferences | Configuration

Overview | Schedule | Activity List | Manage

Online Apps  
Aspen Process Controller  
PETLIN\_ASV2020  
All Variables  
Independents  
Dependents  
OPERATOR  
Aspen IQ  
IQASV2020  
GLOSS  
MFI  
DIEPRESS  
PRODSPEX  
IQPETLIN  
Aspen Process Sequencer  
HPTUBE

Product: C150Y

HPTUBE - C150Y

| Activity Description                             | Action | Start | End   | Detail                                                                       |
|--------------------------------------------------|--------|-------|-------|------------------------------------------------------------------------------|
| 1. Enter DCS Transition Time                     |        | 11:16 | 00:01 |                                                                              |
| 2. Confirm CTA and PEAK APC control              |        | 11:16 | 11:18 | Wait to Change CTA Dosing<br>Time Remaining: 01:25:43<br>End Timer: 13:38:01 |
| 3. Select AUXILIARY for APC Control              |        | 11:18 | 11:19 |                                                                              |
| 4. Confirm AUX target and ramp rate              |        | 11:18 | 11:19 |                                                                              |
| 5. APC Check Controller & Inferential Status     |        | 11:19 | 11:20 | Wait to Change Peak Temp1<br>Time Remaining: 01:50:43<br>End Timer: 14:03:01 |
| ==SECTION A (CTA & PEAK CONTROL)==               |        | 00:00 | 00:01 |                                                                              |
| 6. Time to start CTA Over/Undershoot             |        | 00:00 | 00:01 | Wait to Change Peak Temp2<br>Time Remaining: 01:50:43<br>End Timer: 14:03:01 |
| =6.1 CTA Over/undershoot & MFI Target Change     |        | 00:00 | 00:01 |                                                                              |
| =6.2 [COMPLETE] CTA over/undershoot              |        | 00:00 | 00:01 | Wait to Change Peak Temp3<br>Time Remaining: 01:50:43<br>End Timer: 14:03:01 |
| 7. Time to Peak Temp1 Change                     |        | 00:00 | 00:01 |                                                                              |
| 8. Time to Peak Temp2 Change                     |        | 00:00 | 00:01 | Wait to Change Peak Temp4<br>Time Remaining: 01:50:43<br>End Timer: 14:03:01 |
| 9. Time to Peak Temp3 Change                     |        | 00:00 | 00:01 |                                                                              |
| 10. Time to Peak Temp4 Change                    |        | 00:00 | 00:01 |                                                                              |
| ==SECTION B (AUXILIARY CONTROL)==                |        | 00:00 | 00:01 |                                                                              |
| 11. Time to Product Cooler Temp Change           |        | 00:00 | 00:01 |                                                                              |
| =11.1 [START] Ramping of 12TIC1003.SV            |        | 00:00 | 00:01 |                                                                              |
| =11.2 [COMPLETE] Ramping of 12TIC1003.SV         |        | 00:00 | 00:01 |                                                                              |
| 12. Time to Exit Barrel Water Temp Change        |        | 00:00 | 00:01 |                                                                              |
| =12.1 [START] Ramping of 13TIC0306.RSV           |        | 00:00 | 00:01 |                                                                              |
| =12.2 [COMPLETE] Ramping of 13TIC0306.RSV        |        | 00:00 | 00:01 |                                                                              |
| =12.3 [START] Ramping of 13TIC0313.RSV           |        | 00:00 | 00:01 |                                                                              |
| =12.4 [COMPLETE] Ramping of 13TIC0313.RSV        |        | 00:00 | 00:01 |                                                                              |
| 13. Time to Pelletizer Water Temp Change         |        | 00:00 | 00:01 |                                                                              |
| =13.1 [START] SV Change Pelletizer Water Temp    |        | 00:00 | 00:01 |                                                                              |
| =13.2 [COMPLETE] SV Change Pelletizer Water Temp |        | 00:00 | 00:01 |                                                                              |
| 14. Time to Hot Oil Temp Change                  |        | 00:00 | 00:01 |                                                                              |
| =14.1 [START] SV Change Hot Oil Temp             |        | 00:00 | 00:01 |                                                                              |
| =14.2 [COMPLETE] SV Change Hot Oil Temp          |        | 00:00 | 00:01 |                                                                              |
| 15. [ACKNOWLEDGE] Slip Agent dosing initiated    |        | 00:00 | 00:01 |                                                                              |
| 16. [ACKNOWLEDGE] Carrier dosing initiated       |        | 00:00 | 00:01 |                                                                              |
| 17. [ACKNOWLEDGE] Anti-block dosing initiated    |        | 00:00 | 00:01 |                                                                              |
| 18. [ACKNOWLEDGE] Anti-oxidant dosing initiated  |        | 00:00 | 00:01 |                                                                              |
| ==SECTION C (Quality, check and SS tuning)==     |        | 00:00 | 00:01 |                                                                              |
| 19. Settling time before quality Check           |        | 00:00 | 00:01 |                                                                              |
| 20. APC Check MFI Based Prediction In-Spec       |        | 00:00 | 00:01 |                                                                              |
| 21. APC Check GLOSS Based Prediction In-Spec     |        | 00:00 | 00:01 |                                                                              |
| 22. Wait for settling Time                       |        | 00:00 | 00:01 |                                                                              |
| 23. Load of New Grade APC SS Tuning and Limit    |        | 00:00 | 00:01 |                                                                              |
| ==COMPLETION OF ALL SECTION STATUS==             |        | 00:00 | 00:01 |                                                                              |
| 24. Sect-A (CTA & Peak Adj. Completed)           |        | 00:00 | 00:01 |                                                                              |
| 25. Sect-C (Load of new grade tuning completed)  |        | 00:00 | 00:01 |                                                                              |
| 26. Sect-B (Aux Adj. & Acknowledge Completed)    |        | 00:00 | 00:01 |                                                                              |
| 27. [ACKNOWLEDGE] Transition by GTM APC end      |        | 00:00 | 00:01 |                                                                              |

Update in 15s

ACK/GTM changes Completed  
TimedOut - Waiting for operator response

==COMPLETION OF ALL SECTION STATUS==

24. Sect-A (CTA & Peak Adj. Completed)

25. Sect-C (Load of new grade tuning completed)

26. Sect-B (Aux Adj. & Acknowledge Completed)

27. [ACKNOWLEDGE] Transition by GTM APC end

## Benefits

| <b>N125Y</b>   | <b>Baseline</b> | <b>APC</b> | <b>Improvement</b>    |
|----------------|-----------------|------------|-----------------------|
| Prodspex       | 31.28 T/H       | 31.41 T/H  | + 0.13 T/H            |
| CTA / Prodspex | 13.59 kg/t      | 13.12 kg/t | - 0.47 kg/t           |
| MFI Lab        | 2.53            | 2.41       | Lowered as per target |

| <b>C150Y</b>   | <b>Baseline</b> | <b>APC</b> | <b>Improvement</b>    |
|----------------|-----------------|------------|-----------------------|
| Prodspex       | 31.31 T/H       | 32.33 T/H  | +1.02 T/H             |
| CTA / Prodspex | 17.33 kg/t      | 15.54 kg/t | - 1.79kg / t          |
| MFI Lab        | 5.30            | 5.17       | Lowered as per target |

# PETRONAS

## Passionate about Progress